

Excitation-Aware On-Track System Identification at Full Scale: A Simulation Study

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Abstract—On-track system identification learns tire models from ordinary racing laps instead of dedicated manoeuvres. We study one such residual-learning pipeline at full vehicle scale, in a simulator publishing per-wheel forces as ground truth. Transplanted unchanged from a 1:10 platform it converges on a tire with less than half the vehicle's grip ceiling and most Pacejka coefficients railed. The cause is a loss of excitation created by the pipeline's own virtual-data step, which no change of solver or tuning removes. The friction the synthetic rollout can demand, $\mu_{\text{reach}} = v^2 \delta_{\text{sweep}} / (gL)$, is a property of the rollout configuration and not of the tire, as a grid on a known tire confirms. Two further defects share that source: the fit's objective depends on the coefficients only through the product BCD , so the peak factor is not recoverable from the rollout at any excitation, and the loop feeding each fit its own previous answer is not a contraction. We correct all three (rollout speed from the measured lap, sweep bounded to the observed steering, peak factor measured off the trace and pinned, update damped into a relaxed fixed-point iteration) and gate the controller interface on derived physical quantities rather than on coefficient bounds. Over three timed laps on a surface 33 % below the one its prior was measured on, no accepted coefficient rails, axle cornering stiffness is recovered to 3.2 % front and 1.4 % rear and the peak factor to 11 %, against a railed peak factor and a 118–132 % stiffness error for the inherited configuration. All results are in simulation, one run per configuration.

Index Terms—Autonomous vehicles, system identification, parameter estimation, vehicle dynamics, tires, predictive control, road vehicles, simulation.

I. INTRODUCTION

MODEL-BASED controllers for autonomous racing derive their performance from the fidelity of the vehicle model they carry [1], [2], [3]. In the lateral channel that fidelity is dominated by the tire, simultaneously the physical bound on manoeuvrability and the least accessible component to measure [4], [5]. Classical characterisation takes the vehicle off the track for steady-state circular or ramp-steer manoeuvres, at a logistical cost a race weekend rarely affords; identifying on track removes the logistics but brings back transients, noise and a racing line that is a poor excitation signal, and nonlinear least squares applied directly to lap data is brittle in exactly this regime [6].

Dikici *et al.* [6] propose the hybrid remedy that is the starting point of this paper: a small residual network trained on 30s of ordinary lap data predicts the one-step error of a nominal single-track model; the corrected model is rolled

out through a *synthetic* quasi-steady steering sweep; Pacejka coefficients are fitted to that sweep; and each fit becomes the next nominal model. Because the sweep is generated rather than driven it can be made steady and noise-free, and the residual network only has to interpolate within its training distribution. On a 1:10 platform the procedure reports a $3.3 \times$ lower one-step RMSE than nonlinear least squares under noise [6].

A. Problem statement

This paper asks what the pipeline identifies when it is moved from a 3.7 kg, 0.32 m-wheelbase platform to a full-scale Formula Student vehicle: 269.6 kg, 1.53 m wheelbase, 16° of road-wheel lock, following a reference speed profile of 9.6–25.8 m/s. The question is identifiability, not parameter transfer, and two scale-dependent mechanisms, both latent at 1:10, break the estimate. First, excitation collapses inside the virtual-data step: the Pacejka fit never sees the recorded lap, only the synthetic rollout, whose reachable friction $\mu_{\text{reach}} \approx v^2 \delta_{\text{sweep}} / (gL)$ is a property of the rollout configuration and not of the tire, and a rollout speed inherited from the scaled platform caps it at 0.43 here. Below the tire's peak the Magic Formula degenerates to $F_y \approx F_z(BCD)\alpha$, only the product BCD is identifiable, and the bounded optimiser resolves the remaining freedom on a box edge [7], [4]. Second, lateral demand scales as v^2 , so the tracking cost of a bad grip estimate grows quadratically with speed: one identified set with a peak friction coefficient of 0.40 tracked at 0.16 m of cross-track error at 11.1 m/s and 41.89 m at 27 m/s through the same controller (Section VII-A), and the scaled platform never reaches the speed at which that error becomes observable.

One difference works in our favour. Because the study runs in the CARLA simulator [8] through a purpose-built ROS 2 interface, the simulator's own per-wheel telemetry (slip angle, normal load and lateral force) is available as ground truth, so an on-track identifier can be scored against the forces the plant actually generated.

B. Contributions

- 1) **An identifiability analysis of the virtual-data step** (Section V-B): the closed-form reachable-friction bound above, and an *excitation-aware rollout* that selects the rollout speed from the measured lap and warns when the configuration cannot reach the prior's peak. A grid over rollout speed, sweep amplitude and true peak factor on a known tire both verifies the bound and selects

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the rollout speed floor and sweep-amplitude ceiling the pipeline ships (Section VI-B). This corrects the baseline method [6] at any scale, and stays latent at 1:10 because a 2–4 m/s rollout is representative there and is not here.

- 2) **Two structural properties of the loop, and the corrections they imply** (Section V-D): stage 3’s residual depends on B , C and D only through their product, so the peak factor is not recoverable from the rollout *at any excitation* and must be measured on the recorded trace and held; and the loop is a positive feedback path rather than a contraction, so stage 4 becomes a relaxed fixed-point iteration at $\beta = 0.20$.
- 3) **The corrections the bound implies**, on both sides of the fit (Sections V-C and V-E): a sweep bounded at $1.5 \times$ the 99th percentile of the observed steering, a Tikhonov target frozen at the configured prior, slip angles built from the *achieved* road-wheel angle, mass and wheel-base from measured static wheel loads, a friction warm start from the *curvature* of the axle force curve, and physics-informed dynamics residuals normalised to the state-increment units of the data term, from which raw force and moment residuals sit a factor $2.3e5$ away.
- 4) **Admissibility gates on derived physical quantities** at the model-to-controller boundary (Section VII): axle cornering stiffness, understeer sign and critical speed, and grip ceiling against the trajectory’s lateral demand. They replace per-coefficient box checks, which cannot detect a degenerate fit or a pairwise-oversteering axle set.
- 5) **A simulation architecture that supplies the ground truth, and a closed-loop evaluation on it** (Sections IV-A–IV-D and VI). The evaluation runs on a static 0.70 friction surface, deliberately not the 1.05 the configured tire prior was measured on, so a pipeline unable to move off its prior scores badly. Over three timed laps the corrected pipeline identifies axle cornering stiffness to 3.2 % front and 1.4 % rear and tracks the peak factor to 11 %, against an inherited configuration that rails D on its 2.0 bound in every cycle and overstates stiffness by 118–132 %. Axle force RMSE is $12.1 \times$ and $10.5 \times$ lower, and MPC lateral tracking error 32 % lower, at unchanged solver cost.

All results are obtained in simulation, and Section VIII states which conclusions are expected to transfer to hardware.

II. RELATED WORK AND POSITION

A. On-track and data-driven identification

The Magic Formula [9], [4] is the model identified here. Classical characterisation obtains its coefficients from steady-state manoeuvres in a controlled space [6], [5]: clean data at high logistical cost. Direct fitting to measured contact-patch forces needs a force-measuring hub or a simulator’s own wheel telemetry; here it is used only as a *reference* against which the on-track identifier is scored, never inside the identification loop.

On-track methods trade excitation quality for operational simplicity. Nonlinear least squares on lap data is the classical

baseline and is brittle under noise [6]. The recursive alternative is a dual or joint filter carrying states and parameters together [10], [11], and it is subject to the same excitation condition this paper is about: a filter run on a smooth racing line has as little information about the tire’s peak as a batch fit of the same data, and merely expresses that as a parameter covariance that never shrinks rather than as a coefficient on a bound. Dedicated friction estimators make the excitation dependence explicit [12], [13], [14], [15], and Hsu *et al.* [16] is the closest prior art to the warm start of Section V-E2: they recover the friction limit below the peak from the *curvature* of the force response, read through the pneumatic trail in steering torque. Our estimator uses the same observation on the axle force curve reconstructed from the IMU and odometry, with no steering-torque sensor; it is then gated, applied to both axles from the axle the manoeuvre identifies, and pinned for the co-identification loop, which is a property of what that loop can identify (Section V-D1) rather than of the estimator. Gaussian-process approaches learn the mismatch of a nominal model [17], [18], [2] at a continuing computational cost; neural residuals replace the Gaussian process with a cheaper regressor at the cost of generalisation guarantees.

Deep Dynamics [19] is the closest work on the physics-constraint axis: it estimates Pacejka coefficients inside a neural network and adds a *Physics Guard* layer clamping them into nominal ranges. Our admissibility gates share the motivation but differ in placement. We report as a substantive finding that per-coefficient box constraints are insufficient: the degenerate fit that motivated this work satisfied every box constraint while five of eight coefficients sat exactly on a bound, and a later fit satisfied every box constraint while being pairwise oversteering and open-loop unstable above 14.5 m/s. We therefore gate on derived quantities at the model-to-controller interface and treat a coefficient resting on a bound as a diagnostic of non-identifiability rather than a successful clamp. The hybrid residual-plus-virtual-data structure of [6] is the pipeline we extend; the idea it rests on, to query the learned model only where it interpolates and extract physically parameterised coefficients from it, is retained unchanged, and what we correct is the mechanism deciding *where* the learned model is queried.

B. Overlap with concurrent work

Wu *et al.* [20] extend the same baseline concurrently and independently, introducing a vision-based friction prior, an S4 structured state-space residual in place of the MLP, and Nelder–Mead for the iterative Pacejka extraction. The pipeline described here independently contains an S4D temporal residual [21], [22], a friction warm-start, and Nelder–Mead and differential-evolution solver options, so we claim none of the three as contributions. Three differences are technical rather than chronological. Our friction prior is proprioceptive, so it needs no camera and no textured-surface assumption. It is read from the *curvature* of the axle force curve by a gated brush-model fit [23], [14] rather than from utilised friction, which on a gentle lap measures 0.044 median on this vehicle and is only a lower bound (Section V-E2a). And it is *pinned* rather

than used as an initialisation, because Section V-D1 shows the rollout fit cannot separate the peak factor from the cornering stiffness at any excitation, so a prior on D the fit is free to move is overwritten within one cycle. Neither prior work is the fix: the degeneracy is located in the rollout excitation, not the initialisation, and to our knowledge the reachable-friction bound of Section V-B and its consequence for the virtual-data step have not been stated before.

C. Simulation platform and downstream consumer

CARLA [8] supplies the sensor and scenario infrastructure used here. Its native dynamics come from the underlying PhysX wheeled-vehicle model [24], a saturating friction-and-stiffness tire law (Section IV-B) rather than an empirical Magic Formula fit, which is why this study treats the simulator as a well-instrumented *plant* to be identified and not as a source of ground-truth Pacejka coefficients. R-CARLA [25] makes the dynamics model interchangeable, and Sagmeister *et al.* [26] find that a controller's sensitivity to model fidelity is largest near the acceleration limits, where this study operates; CARLA has also hosted complete Formula Student Driverless stacks with ROS 2 interfaces [27]. Scaled platforms, FITENTH [28] and the ForzaETH Race Stack [3], provide the software context of the baseline; full-scale Formula Student Driverless systems [5] the vehicle class targeted here. The identified model is consumed by a nonlinear MPC path-tracking controller built on *acados* [29] with partial-condensing HPIPM [30] [1], [2], tracking a minimum-curvature raceline [31]. We claim no contribution to the controller; it appears only as the consumer that defines what an admissible identification is (Section VII).

III. IDENTIFICATION PROBLEM

A. Model structure

The identification model is the dynamic single-track model of the baseline, with mass m , yaw inertia I_z and centre-of-gravity distances $l_f, l_r, L = l_f + l_r$. Longitudinal and lateral dynamics are decoupled and load transfer is neglected:

$$\dot{v}_y = \frac{1}{m}(F_{y,r} + F_{y,f} \cos \delta - mv_x \omega), \quad (1)$$

$$\dot{\omega} = \frac{1}{I_z}(F_{y,f} l_f \cos \delta - F_{y,r} l_r), \quad (2)$$

with slip angles

$$\alpha_f = \delta - \arctan\left(\frac{v_y + l_f \omega}{v_x}\right), \quad \alpha_r = -\arctan\left(\frac{v_y - l_r \omega}{v_x}\right). \quad (3)$$

Axle forces follow the lateral Magic Formula

$$F_y = F_z D \sin(C \arctan(B\alpha - E(B\alpha - \arctan B\alpha))), \quad (4)$$

with per-axle coefficients $\Phi_p = [B_f, C_f, D_f, E_f, B_r, C_r, D_r, E_r]$. The identification state and input are $\mathbf{x} = [v_y, \omega]$ and $\mathbf{u} = [v_x, \delta]$, and the nominal one-step map is $\hat{\mathbf{x}}_{k+1} = f(\mathbf{x}_k, \mathbf{u}_k, \Phi_p)$. The baseline discretises (1)–(2) by explicit Euler at T_s ; the integrator is selectable and the shipped configuration uses RK4, because the full-scale vehicle's lateral mode is stiffer relative to T_s than the scaled platform's. The inherited arm of Section VI keeps Euler.

B. Objective and derived quantities

The objective is to recover Φ_p from a single 30 s segment of ordinary lap data, without dedicated manoeuvres, such that the resulting model is both predictive and *admissible* for the controller that consumes it. Admissibility is expressed on derived quantities rather than on coefficients. The linearised axle cornering stiffness is the slope of (4) at zero slip,

$$C_i = F_{z,i} B_i C_i^{\text{MF}} D_i, \quad i \in \{f, r\}, \quad (5)$$

(writing C^{MF} for the shape factor to avoid a clash), and the grip ceiling implied by an identified set is

$$a_y^{\text{max}} = \frac{D_f F_{z,f} + D_r F_{z,r}}{m}. \quad (6)$$

The linear model built from (5) is oversteering when $l_f C_f > l_r C_r$ and is then open-loop unstable above the critical speed $v_{\text{crit}} = L \sqrt{C_f C_r / (m(l_f C_f - l_r C_r))}$ [32].

C. Practical identifiability under limited excitation

Two properties of (4) are used throughout. Its slope at zero slip is $F_z BCD$, so the cornering stiffness depends on the coefficients only through that product, and E drops out. And if the data never approaches the peak, (4) linearises to $F_y \approx F_z (BCD)\alpha$, leaving B, C and D mutually unidentifiable. Given data that does reach the peak, (4) is *structurally* identifiable [33]; what fails here is *practical* identifiability [34], the standard persistent-excitation condition [7], [35] written on this model, and we claim neither result as novel. Our claim, demonstrated in Section V-B, is that the baseline pipeline's virtual-data step can silently violate that condition, and that a bounded optimiser then resolves the remaining freedom by sliding onto a box edge. The violation surfaces as a converged fit rather than a failure.

D. Baseline pipeline

The baseline repeats four stages for N_{it} iterations [6]. A small network is trained on the recorded lap to predict the nominal model's one-step error from $[v_x, v_y, \omega, \delta]_k$ (stage 1); the corrected model is integrated open-loop at constant speed while the steering angle is ramped from 0 to δ_{sweep} , producing a synthetic quasi-steady dataset (stage 2); assuming $\dot{v}_y = \dot{\omega} = 0$ on that data, axle forces are recovered from the yaw balance,

$$F_{y,r} = \frac{ml_f v_x \omega}{L}, \quad F_{y,f} = \frac{ml_r v_x \omega}{L \cos \delta}, \quad (7)$$

and Φ_p is fitted to $(\alpha_i, F_{y,i})$ by bounded least squares (stage 3); the fitted Φ_p then becomes the next iteration's nominal model (stage 4). Algorithm 1 gives the corrected cycle in full.

The rest of this paper follows from one property of that structure: stage 3 never sees the recorded lap, only the output of stage 2. The recorded lap enters only through the residual network, so the excitation available to the fit is set by the rollout configuration and not by how the vehicle was driven.

TABLE I
VEHICLE CONFIGURATION AND MEASURED PLANT PROPERTIES.

Quantity	Value	Unit
Body mass (configured)	240.0	kg
Static wheel load (total)	2644.5	N
Total (kerb) mass m^a	269.6	kg
Front/rear static split	51.2 / 48.8	%
Wheelbase L	1.5334	m
l_f / l_r	0.7381 / 0.7953	m
Yaw inertia I_z^c	51.1	kg m ²
Max road-wheel angle	16.0	°
Configured wheel friction ^d	1.0	—
Road-surface factor	0.70	—
Effective friction ^e	0.70	—
Realised peak axle friction	0.71	—

^aTotal static wheel load divided by g ; this is the mass every equation in this paper uses, not the configured body mass. ^cAssumed; see Section A. ^dA scenario parameter, not a vehicle property. ^eConfigured wheel friction multiplied by the road surface's own coefficient, as reported per wheel. It is the effective value, not the configured one, that the identifier is scored against (Section V-E2a).

IV. SIMULATION ARCHITECTURE AND DATA PIPELINE

The pipeline is exercised end to end inside a single architecture: a Formula Student vehicle model in CARLA acting as the plant, a C++ ROS 2 communication layer that drives it and republishes its state, and an identification stack that consumes that state. CARLA [8] was selected because it exposes per-wheel slip angle, normal load and tire force rather than chassis states alone, and because its synchronous mode with a fixed physics step (0.033 s here) stamps every sample on the simulation clock; those two properties are what make the identifiability claims of Section V checkable rather than arguable [36].

A. Plant: Formula AI vehicle in CARLA

The plant is the driverless Formula Student-class vehicle model, instantiated in CARLA (server 0.9.16, Unreal Engine 4.26) on the *silverstone* map: a four-wheeled, front-steered, all-wheel-drive rigid-body vehicle whose lateral forces are generated by the PhysX wheeled-vehicle tire model [24] under a configured friction coefficient and lateral stiffness, not by a Magic Formula. The simulator therefore has no “true” Pacejka coefficients to recover; instead the lateral forces PhysX computes at each wheel are taken as ground truth, and the identified Magic Formula is validated against those forces.

B. The plant's tire curve, in closed form

The identification target is known here in closed form. PhysX computes the lateral force of each wheel with `PxVehicleComputeTireForceDefault` [24]; at zero longitudinal slip and zero camber that expression collapses to

$$F_y(\alpha) = \text{sgn}(\alpha) \mu F_z S_1(K), \quad K = \frac{C_{\text{lat}} |\tan \alpha|}{\mu F_z}, \quad (8)$$

with $C_{\text{lat}} = F_z^{\text{rest}} k_y S_1(3\bar{F}_z/k_x)$, $\bar{F}_z = F_z/F_z^{\text{rest}}$ the normalised load, (k_y, k_x) the wheel's lateral-stiffness parameters, and $S_1(K) = \min(1, K - K^2/3 + K^3/27)$ the engine's smoothing function.

TABLE II
THE PLANT'S LATERAL TIRE CURVE: THE PARAMETERS OF (8), READ FROM THE SIMULATOR'S OWN WHEEL CONFIGURATION AND TELEMETRY.

Quantity	Value	Unit
Effective friction μ	0.70	—
Lateral stiffness k_y	17.0	/rad
Max normalised load k_x	2.0	—
Initial slope C_{lat} at $\bar{F}_z = 1$	14.875 F_z^{rest}	N/rad
front axle, $F_{z,f} = 1373$ N	20.4	kN/rad
rear axle, $F_{z,r} = 1274$ N	19.0	kN/rad
Peak reached at, $\bar{F}_z = 1, \mu = 0.70$	8.0	°

The initial slope is set by the *rest* load and by the normalised load through S_1 , not by F_z alone, and falls monotonically with $\bar{F}_z$. Every axle slope quoted in this paper is taken at the static axle loads listed here; the closed-form values in this table are the $\bar{F}_z = 1$ ones, and a slope fitted to telemetry from a run that loads the outside wheels beyond the rest load comes out below them: 17.4 kN/rad front on the $\mu = 0.70$ run of Section VI-D, against 20.4 kN/rad here.

The peak angle follows from $K = 3$ in (8) and therefore moves with μ : 8.0° at the effective $\mu = 0.70$ of the runs reported here, 12.0° at the $\mu = 1.05$ of the prior-fitting run.

Two properties of (8) matter downstream. It saturates at μF_z , first reached at $K = 3$, and it never falls off past that peak, so a Magic Formula can match it up to the peak and no further: E and the post-peak decay exist to describe a falling branch this curve does not have. Its scale is set entirely by μ and C_{lat} , both of which the simulator publishes, so the curve of Table II is read off the plant rather than fitted to it.

That the closed form is the right one was checked rather than assumed: on 1804 ground-truth wheel samples from a live lap ($P_{99}(|\alpha|) = 3.66^\circ$), each predicted from its own measured α , F_z and μ , (8) returns $R^2 = 0.99$ at 16.1 N front and 14.7 N rear RMSE. Section VI-D uses this curve to separate the error of the identified fit from the error of representing (8) in the form (4) that the controller requires.

C. CARLA_ASU_BRIDGE: interfaces and telemetry

CARLA_ASU_BRIDGE [37] is a C++ ROS 2 lifecycle node linked directly against CARLA's client library (Fig. 1). It spawns and configures the ego vehicle, drives its inputs over a single `ackermann_msgs/AckermannDriveStamped` on `/drive`, and republishes simulator state as standard ROS 2 topics; all identification traffic crosses this interface. On the feedback path, one property is specific to this architecture and the rest of this paper rests on it: the simulator's per-wheel slip angle, normal load and lateral force are republished on the same simulation clock as the odometry the identifier consumes, so an identified tire model can be scored against the forces the plant actually generated rather than a second model. Appendix B lists the full interface and the body-frame and sign conventions at the simulator-ROS 2 boundary.

D. Data acquisition and preprocessing

A model-free pure-pursuit controller drives the vehicle around the reference raceline; it requires no tire knowledge beyond the wheelbase and is therefore a valid bootstrap, as in the baseline [6], [3]. Odometry, the achieved steering angle and the drive command are sampled into a ring buffer by a 50 Hz timer for 30 s, gated on $v_x > 1$ m/s. The timer runs

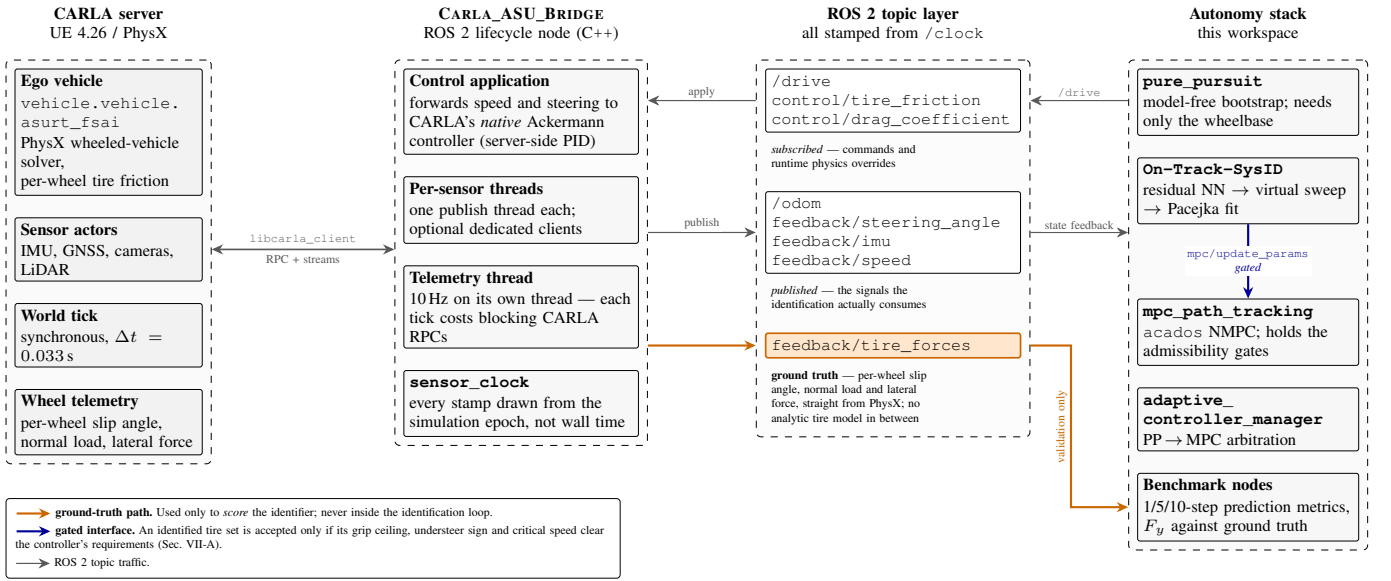


Fig. 1. Architecture of the CARLA_ASU_BRIDGE ROS 2 interface. The bridge links against CARLA's C++ client, drives the ego vehicle through CARLA's native Ackermann controller, and republishes state, sensors and per-wheel telemetry on ROS 2 topics stamped from the simulation clock. The *feedback/tire_forces* path (highlighted) carries the simulator's own per-wheel slip angle, normal load and lateral force, and is the ground truth against which the identifier is scored.

on the simulation clock, which advances only at the server's 0.033 s step, so the achieved acquisition rate is the step rate.

Two mismatches follow. The acquisition timer asks for a sample every 0.020 s, 65 % more often than the server produces one, so a fraction of consecutive buffer entries are repeated states and the one-step residual target carries a zero-order-hold artefact on those pairs. Separately, the one-step model of Section III-A is discretised at $T_s = 0.035$ s, 6 % longer than the physics step. T_s is empirical and is carried because the identification proved insensitive to it: at 0.020 s and 0.035 s the corrected configuration returns the *same* last accepted coefficient set to four decimal places, and the inherited configuration keeps D on its 2.0 bound at both, so the box-edge signature is not an artefact of the mismatch. Results are reported at $T_s = 0.035$ s. Preprocessing follows the baseline: a zero-phase low-pass filter, then symmetry augmentation by mirroring $(v_y, \omega, \delta) \rightarrow (-v_y, -\omega, -\delta)$ at unchanged v_x . Re-identification retriggers every 30 s by default.

Only the deployment stage leaves this architecture: validation draws on *feedback/tire_forces* and *feedback/imu*, neither of which enters the identification loop, so the estimator is scored against a channel it did not observe, and Section VI can attribute a closed-loop difference to the identified model rather than to the interface carrying it.

V. IDENTIFICATION METHODOLOGY

Figure 2 summarises the loop and Algorithm 1 states it in the form needed to reimplement it; lines marked $\triangleright$ are this paper's changes to the inherited pipeline. Sections V-B–V-C carry the central methodological contribution.

in: buffer $\mathcal{D} = \{v_x, v_y, \omega, \delta^{\text{ach}}\}_{1:K}$ (30 s), prior Φ_p^0 , bounds $[\Phi_p^-, \Phi_p^+]$
out: Φ_p , or UNIDENTIFIED

- 1: $\delta \leftarrow$ achieved road-wheel angle, command on staleness $\triangleright$
- 2: $m, l_f, l_r \leftarrow$ measured static wheel loads $\triangleright$
- 3: $\hat{\mu}_f \leftarrow \text{BRUSHFIT}(\mathcal{D})$, front axle, released only if gated $\triangleright$
- 4: **if** released **then** $D_f^0, D_r^0 \leftarrow \hat{\mu}_f$; D held for the whole loop $\triangleright$
- 5: $\Phi_p^{\text{tik}} \leftarrow \Phi_p^0$ (frozen for all iterations) $\triangleright$
- 6: $v \leftarrow \text{clip}(\bar{v}_x(\mathcal{D}), v_{\min}, \infty)$ $\triangleright$
- 7: $\delta_{\text{sweep}} \leftarrow \text{clip}(1.5 P_{99}(|\delta|), \delta_{\min}, \delta_{\max})$ $\triangleright$
- 8: $\mu_{\text{reach}} \leftarrow v^2 \delta_{\text{sweep}} / (gL)$; **warn if** $\mu_{\text{reach}} < \max_a D_a^0$ $\triangleright$
- 9: $\Phi_p \leftarrow \Phi_p^0$; **for** $i = 1$ **to** N_{it} **do** $\triangleright$
- 10: $f_{NN} \leftarrow \text{TRAINRESIDUAL}(\mathcal{D}, \Phi_p)$; $S \leftarrow \text{ROLLOUT}(\Phi_p, f_{NN}, v, \delta_{\text{sweep}})$ $\triangleright$
- 11: $\Phi_p \leftarrow \arg \min_{B, C, E} \|F_y(\Phi_p; S) - \hat{F}_y(S)\|^2 + \lambda \|\Phi_p - \Phi_p^{\text{tik}}\|^2$, D held $\triangleright$
- 12: $\Phi_p \leftarrow (1 - \beta) \Phi_p + \beta \tilde{\Phi}_p$ ($\beta = 0.20$) $\triangleright$
- 13: **if** any Φ_p component on a bound **then** report the degeneracy $\triangleright$
- 14: **if** not GATES(Φ_p) **then** return UNIDENTIFIED $\triangleright$
- 15: submit Φ_p ; keep the previous model until acknowledged $\triangleright$

: GATES: axle stiffness sign and magnitude; $l_f C_f < l_r C_r$ and v_{crit} outside the speed range; grip ceiling $\geq$ the trajectory's lateral demand (Section VII).

Algorithm 1: Corrected on-track identification cycle. $\triangleright$ marks a change to the inherited pipeline [6].

A. Residual model and objective

The residual carries whatever the nominal model (1)–(4) does not represent. The default is the baseline multilayer perceptron: 4 inputs, one hidden layer of 8 units, 2 outputs, LeakyReLU, 58 parameters, full-batch Adam [38] at 5×10^{-4} on an MSE objective, in PyTorch [39]. The architecture is selectable; the two arms of Section VI use baseline and s_4 , the latter an S4D diagonal state-space residual [21] over a sliding window of $W = 20$ samples (0.7 s), so the residual can carry tire-relaxation and yaw-damping dynamics a per-sample map cannot. We claim neither that architecture, which is also in concurrent work [20], nor the $8.7\times$ reduction in

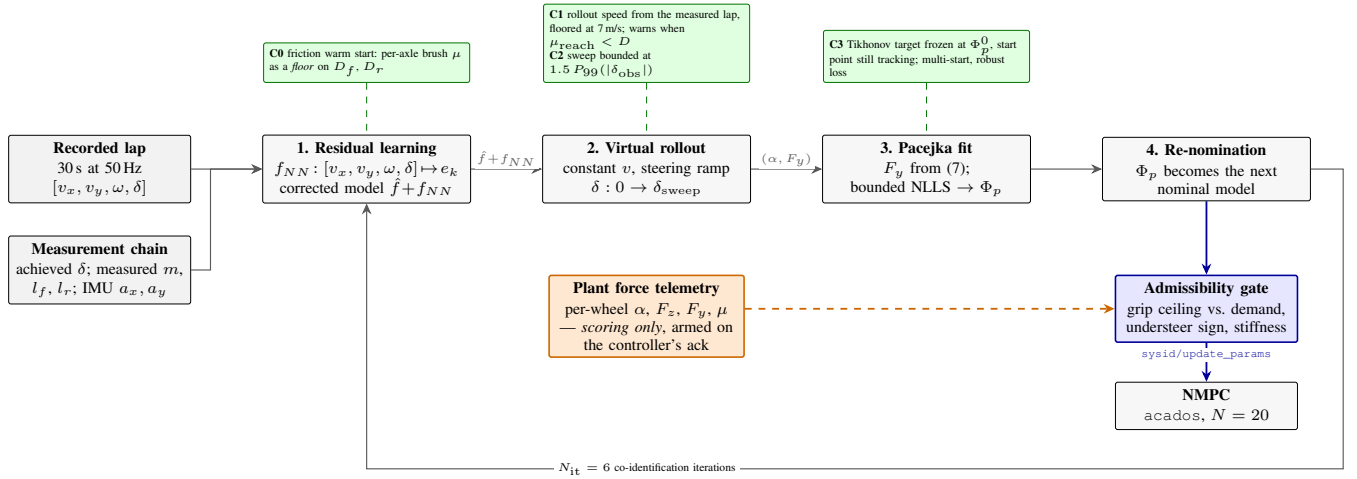


Fig. 2. The identification pipeline. Recorded lap data trains the residual network, the corrected model is rolled out through a synthetic steering sweep, Pacejka coefficients are fitted to that sweep, and the fit becomes the next nominal model under relaxation. Green marks this paper's changes: the friction warm start (Section V-E2), the excitation-aware rollout speed and extrapolation-bounded sweep (Sections V-B–V-C), the frozen regularisation target of (12) and the relaxed update of (14). Blue is the gated controller interface (Section VII); orange is the plant's per-wheel force telemetry, which scores the result and never enters the loop.

its training cost, an exact algebraic rewrite that changes no identified coefficient; both are in the supplementary material, and the identifiability result below holds for either architecture.

The input space of the `s4` arm is augmented with the axle slip angles of (3): since the tire law (4) is a function of (α, F_z) and of nothing else, supplying α_f, α_r directly spares the network a relation the model already states. The augmentation is also available on the MLP, at 16 extra weights ($58 \rightarrow 74$ parameters), and is off in the `baseline` arm. The optional physics-informed objective

$$\mathcal{L} = \mathcal{L}_{\text{data}} + \lambda_s \mathcal{L}_{\text{steady}} + \lambda_m \mathcal{L}_{\text{sym}} + \lambda_g \mathcal{L}_{\text{smooth}}, \quad (9)$$

with $\lambda_s = 0.1$, $\lambda_m = 0.05$, $\lambda_g = 10^{-4}$, adds three penalties to the one-step MSE [40], [19]: $\mathcal{L}_{\text{steady}}$ keeps the two residual channels jointly consistent with (1)–(2) on a quasi-steady mask, so they cannot imply an axle force pair no tire could produce (stage 3 fits that force pair, not the state prediction); $\mathcal{L}_{\text{sym}}$ enforces the odd symmetry $e(-\mathbf{z}) = -e(\mathbf{z})$ of a laterally symmetric vehicle; $\mathcal{L}_{\text{smooth}}$ is a soft Lipschitz bound.

One property of (9) is not a hyperparameter. $\mathcal{L}_{\text{data}}$ is a mean square over a one-step increment while the two balance residuals are a force and a moment, four to six orders of magnitude apart, so λ_s is not a relative weight at all: measured at initialisation on a 30 s buffer of this vehicle, the unnormalised steady term contributes 99.9996 % of $\mathcal{L}$ against 0.0004 % for the data term, a ratio of $2.3e5$. Such a residual fits the nominal model's own force imbalance rather than the data it was given. Each residual is therefore expressed as the state increment it implies, the unit the network's output is in,

$$r_{\text{lat}} = \frac{T_s}{m} \left(m\dot{v}_y + mv_x\omega - (F_{y,r} + F_{y,f} \cos \delta) \right), \quad (10)$$

and r_{yaw} likewise with T_s/I_z . The shares become 98.53 % data and 1.47 % steady, recovering the meaning (9) gives λ_s . The weights in (9) are relative weights only in these units.

B. Excitation-aware virtual rollout

Consider stage 2 of Section III-D: a constant-speed rollout at v with the steering ramped to δ_{sweep} . In quasi-steady cornering the lateral acceleration a single-track vehicle develops kinematically is $a_y = v^2 \kappa$ with $\kappa \approx \delta/L$, so the largest friction coefficient the rollout can demand of the tire is

$$\mu_{\text{reach}} \approx \frac{v^2 \delta_{\text{sweep}}}{g L} \quad (11)$$

Equation (11) contains no tire parameter: it is a property of the rollout configuration alone, and an *upper* bound rather than an equality, since the steady-state curvature a compliant vehicle holds is smaller than the Ackermann δ/L . Reading it as a ceiling is all the argument needs: if even the ceiling sits below D , the synthetic data never leaves the linear ramp of (4), only the product BCD is identifiable (Section III-C), and the bounded optimiser resolves the residual freedom on a box edge.

The baseline implementation selects the rollout speed as $v = \text{clip}(\bar{v}_x, 2, 4)$ m/s, representative of a 1:10 platform. On our vehicle, with $\delta_{\text{sweep}} = 0.4$ rad and $L = 1.5334$ m, (11) gives $\mu_{\text{reach}} = 0.43$ at 4 m/s: the rollout cannot ask the tire for more than 0.43 g whatever the tire is. At 1:10 scale, with $L \approx 0.32$ m, the same speed yields $\mu_{\text{reach}} \approx 2.0$ and the pathology is invisible. The rollout speed is therefore now taken from the speed the vehicle was actually driven at, clamped below at 15 m/s, and the pipeline warns whenever (11) evaluates below the prior's D , turning a silent degeneracy into a startup diagnostic. The same bound sets a data-collection requirement: identification data should be gathered at $v \geq \sqrt{gLD/\delta_{\text{sweep}}}$.

C. Bounded extrapolation and frozen-prior regularisation

Raising the rollout speed exposes a second, opposite failure: the rollout queries the residual network at steering angles it has never seen, the observed $|\delta|$ on a pure-pursuit lap staying below ≈ 0.06 rad while the sweep ran to 0.4 rad.

The fit reads the network's extrapolation as additional grip, and because each iteration's answer became both the next iteration's nominal model *and* its regularisation target, the error compounds: D walked from the 1.009/1.002 prior to 1.5402/1.8721 over six iterations, a claimed 1.7 g on a vehicle measuring 1.0 g in steady state.

Two changes close the loop. First, an extrapolation bound: the sweep terminates at $\delta_{\text{sweep}} = \text{clip}(\lambda_e P_{99}(|\delta_{\text{obs}}|), \delta_{\text{min}}, \delta_{\text{max}})$ with $\lambda_e = 1.5$, $\delta_{\text{min}} = 0.05$ rad and $\delta_{\text{max}} = 0.54$ rad, so the network is never queried more than 50 % beyond the steering range it was trained on. Both δ_{max} and the inherited 0.4 rad lie above the plant's 0.279 rad mechanical lock deliberately: the sweep is a *virtual* rollout, and clamping it to the lock would cap μ_{reach} a second time, by the actuator instead of the rollout speed (Section A, A6). Second, a frozen prior: the Tikhonov target is held at the configured Φ_p^0 for all iterations while the optimiser's *start point* continues to track the previous iteration, giving the per-axle objective

$$\min_{\phi} \sum_k \rho(r_k(\phi)) + \sum_j w_j (\phi_j - \phi_j^0)^2, \quad (12)$$

with r_k the force residual, ρ a soft- ℓ_1 robust loss and w_j scaled to the data residual budget by a single dimensionless weight $\lambda = 0.05$. The regulariser exists to hold the coefficients the manoeuvre does not excite, E above all, at their nominal value instead of on a box edge.

D. What the rollout fit identifies, and why the loop must be damped

1) *The rollout fit sees only the product BCD*: In the virtual-data step the low-slip degeneracy of Section III-C is stronger, and structural rather than excitation-dependent. Stage 3 does not observe a force; it reconstructs one from the rollout's own quasi-steady yaw balance (7), which on the synthetic trajectory reduces to

$$\frac{F_{y,f}}{F_{z,f}} = \frac{v_x \omega}{g \cos \delta}, \quad \frac{F_{y,r}}{F_{z,r}} = \frac{v_x \omega}{g}, \quad (13)$$

so the only quantity (12) ever compares against is the lateral acceleration the rollout realises, which the *previous* coefficient set produced through its slope $F_z BCD$. The fit's residual therefore depends on (B, C, D) only through their product, and D is the free coordinate of the hyperbola $BCD = \text{const}$. Replayed on five identification buffers of a live $\mu = 1.05$ run, the product is recovered tightly, $BCDF_{z,f}$ over 18.9–21.1 kN/rad against the 20.4 kN/rad the plant carries at $\bar{F}_z = 1$ (Table II; that run stays below the peak, so the fitted and closed-form slopes agree there), while the D reaching that same product ranges over 0.69–2.00. D must therefore not be asked of this fit at all. It is measured on the *recorded* trace by the brush estimator of Section V-E2, which fits $F_y(\alpha)$ directly and is not subject to (13), then held fixed while B , C and E carry the stiffness the rollout does determine.

2) *The loop is not a contraction*: Stage 4 makes each iteration's answer the nominal model stage 2 rolls out, and stage 3 then fits the trajectory that model produced, over a steering range already extrapolating $1.5 \times$ past the observed

one. The fit reads its own forces back, so the loop is a positive feedback path on its own extrapolation error rather than a contraction: replayed offline on five recorded buffers, axle force RMSE against the plant's telemetry degrades monotonically with iteration count, 38.9, 53.0, 58.5 and 70.4 N at the front for $N_{\text{it}} \in \{1, 2, 3, 6\}$. Pinning D does not remove this; it relocates it into B , C and E . The remedy is to damp the step, not to shorten the loop, so stage 4 becomes a relaxed fixed-point iteration

$$\Phi_p^{(i)} \leftarrow (1 - \beta) \Phi_p^{(i-1)} + \beta \tilde{\Phi}_p^{(i)}, \quad (14)$$

with $\beta = 0.20$; $\beta = 1$ recovers the inherited update exactly. The sweep over β (Section VI-I) carries the control the contribution needs: $\beta = 0$ freezes the coefficients at the pinned prior, performing no identification at all, and is the worst setting. The iterations do carry information, but only a damped update preserves it.

Fitting itself applies (7) to the synthetic data under box constraints $\Phi_p \in ([4.0, 1.2, 0.4, -3.0], [20.0, 2.2, 2.0, 1.0])$ per axle, over B , C and E with D held. Relative to the baseline we raised the jittered multi-starts from 1 to 8 and added a derivative-free simplex [41] and a global differential-evolution solver [42] alongside trust-region reflective [43], through the same library [44]. Neither change addresses the degeneracy, and Section VI-B verifies both as secondary.

E. Measurement-side corrections

1) *Achieved steering, measured mass and geometry*: α_f in (3) was built from the /drive setpoint. The measured achieved/commanded ratio is 0.76, so α_f was systematically $\approx 25\%$ too large and the fitted front stiffness $\approx 20\%$ too small (13725 N/rad against 17198 N/rad fitted to the simulator's own per-wheel forces over the same run); using `feedback/steering_angle` recovers 18486 N/rad. Separately, m and l_{wb} come from the measured static wheel loads (Table I) rather than a configuration file, and $l_{wb} \equiv l_f + l_r$ is enforced by a regression test, because (7) uses the wheelbase for the normal loads and the axle split for the force division, and the two disagreeing is a silent bias.

2) *Proprioceptive friction warm-start*: The inherited pipeline had no prior for the peak factor: D started from a configuration constant and was left to the fit. Newton–Euler on the single-track body of (1)–(2), solved for the axle forces rather than the accelerations, gives

$$F_{y,f} = \frac{I_z \dot{\omega} + l_r m a_y}{L \cos \delta}, \quad F_{y,r} = \frac{l_f m a_y - I_z \dot{\omega}}{L}, \quad (15)$$

which needs the IMU's lateral acceleration and the yaw acceleration and no tire model; vertical loads carry the longitudinal transfer $F_{z,f/r} = m(g l_{r/f} \mp a_x h_{cg})/L$ [32], and yaw acceleration is the derivative of a local polynomial fit to the yaw rate (Savitzky–Golay, window 21, order 2) [45] rather than a difference, because $I_z \dot{\omega}$ is a $\approx 10\%$ term in $F_{y,f}$ here.

Per axle, cornering stiffness and peak friction are then fitted jointly to the Fiala brush model [23], [46], [4]

$$F_y = \mu F_z (1 - (1 - \xi)^3) \text{sgn } \alpha, \quad \xi = \min\left(\frac{C_\alpha |\tan \alpha|}{3\mu F_z}, 1\right), \quad (16)$$

by bounded nonlinear least squares with a soft- ℓ_1 loss [44], [43], over samples with $|a_x| \leq 0.35 g$, the fit being pure-lateral while combined slip shrinks the lateral capacity [14]. The brush model replaces (4) here because it has two parameters, both of which the manoeuvre can support, against the four of the model whose non-identifiability at low slip this paper is about. What beats the utilisation bound is that μ enters (16) through the *curvature* of $F_y(\alpha)$ and not only through its asymptote: writing $r = |F_y|/(C_\alpha |\tan \alpha|)$ for the force's departure from its own linear extrapolation, (16) inverts to

$$\mu = \frac{2k}{3 - \sqrt{3(4r - 1)}}, \quad k = \frac{C_\alpha |\tan \alpha|}{3F_z}, \quad (17)$$

so a 30 % departure from linearity ($r = 0.7$) already pins μ , well below the limit.

Equation (17) also states its own failure mode: $d\mu/dr$ grows without bound as the curve stays linear. The estimate is released only when four conditions hold: absolute grip demand $\max |F_y|/F_z \geq 0.40$, peak friction utilisation $\max |F_y|/(\hat{\mu}F_z) \geq 0.40$, relative precision $\sigma_{\hat{\mu}}/\hat{\mu} \leq 0.15$ from the Gauss–Newton covariance, and $\hat{\mu}$ not resting on its own search bound, $[0.2, 2.0]$; otherwise the module logs which condition failed and falls back to the utilisation bound. The first two are the same quantity with and without the fit in the denominator, and both are needed: the utilisation test divides by the *fitted* $\hat{\mu}$ of the same least-squares call, so an axle whose μ comes out too low clears the gate on its own error, while the absolute ratio has no such freedom. The 40 % floor is the level the friction-estimation literature converges on [14].

a) Applied two-sided, then pinned: The released $\hat{\mu}$ replaces D on both axles, in either direction, and D is then held fixed for the whole co-identification loop. Both axles take the *front* fit: the rear slip angle does not carry δ , so on this trajectory the rear force curve has too little curvature to support (17) and its μ comes out high (1.495/1.383/1.257 rear against 1.011/0.964/0.837 front on three admitted buffers of a $\mu = 1.05$ run). This is not a yaw-inertia error: in quasi-steady cornering $F_{y,f}/F_{y,r} \rightarrow l_r/l_f$, the same ratio as $F_{z,f}/F_{z,r}$, so $\mu_f = \mu_r$ whatever I_z is, and there is one road surface. It is two-sided rather than a floor, the rule it replaces having protected against a D that *moved* between cycles rather than one that was low; it is pinned and not merely seeded, since (13) leaves D free along $BCD = \text{const}$ and on a live run a 0.785 seed ended at the 2.000 bound within one cycle; and an axle with no released $\hat{\mu}$ holds the D it came in with, since there is nothing in the rollout to fit it against.

Pinning puts the whole weight of the result on the gate, and σ_μ/μ from the fit covariance is a *precision* figure, not an accuracy one: a wrong mass, yaw inertia, IMU bias or road bank all fit tightly and wrongly through (15). Here the configuration file's 240.0 kg against the 269.6 kg the wheels actually carry biased μ 11 % low until the measured value replaced it, and the configured wheel coefficient is not the effective one the physics step uses, the server multiplying it by the road surface's 0.70.

F. Deployment and validation protocol

The identifier runs as a single ROS 2 node cycling a rolling FIFO buffer through four states (collect, train, await the controller's acknowledgement, run), re-identifying every $T_{\text{reid}} = 30$ s. Coefficients are submitted over a service call whose non-acknowledgement leaves the previous model in place, so a rejected identification degrades to the last accepted model rather than to none. That acknowledgement also arms the scoring: a submission the controller *rejects* on its admissibility gate (Section VII) is a model the car never drove on, and the seconds between a fit completing and its acknowledgement landing are seconds still driven on the previous model.

Three levels of validation are used, in increasing strength, and all three score the *accepted* model only. One-step prediction on a held-out run, as in the baseline: RMSE and R^2 of v_y and ω , with online statistics in Welford's form [47] and samples admitted only above 1 m/s. Direct force comparison against `feedback/tire_forces`: the identified curve is evaluated at the measured (α, F_z) and compared per axle against the measured F_y , and against the plant's own curve (8) at the same points, which separates the error of the fit from the error of representing a PhysX tire. Closed-loop tracking under the MPC: lateral deviation, its time-weighted integral, heading error and solver cost, split by active controller so the pure-pursuit acquisition phase is not scored against a model it does not use.

VI. EXPERIMENTS AND RESULTS

A. Setup

All experiments run on the architecture of Section IV: the Formula AI vehicle in CARLA driven through ROS 2 Humble, the server in synchronous mode at a 0.033 s fixed physics step. The identifier collects 30 s of data on a 50 Hz timer that the server's step rate holds to 30 Hz (Section IV-D) and re-identifies from the newest buffer, matching the baseline's data budget [6]. Ground truth is the simulator's own per-wheel telemetry, which the identifier never reads; every figure is exported with the CSV holding its samples, and the cross-run comparison reads those CSVs rather than re-deriving any metric.

Two clocks appear below, and no metric mixes them. The controller node stamps its own quantities (run duration, handover time, lap time and every tracking error) on the ROS wall clock, on which the two runs last 580.4 s and 554.6 s. The identifier and the force benchmark stamp theirs on the simulation clock, which the server advances faster than wall time (1.56 and 1.66 here) and on which the same two runs last 906.6 s and 920.7 s.

1) Surface: Every run uses a *static* surface: the per-wheel coefficient is held at its configured 1.0, and the road-surface factor of 0.70 PhysX applies on top of it makes the effective coefficient 0.70, constant to the last digit the simulator reports over all 10744 and 12025 force-validation samples of the two runs. It is deliberately *not* the $\mu = 1.05$ surface the tire prior was measured on (Section VI-A4), so an arm that cannot move off its prior cannot score well here. It remains a *single* surface:

TABLE III
THE TWO CONFIGURATIONS.

Setting	Baseline-NN-MSE	Excitation-S4D-Pinn
Residual architecture	MLP (baseline)	S4D (s4)
Residual objective	One-step MSE	Physics-informed, (9)
Model integrator	Euler	RK4
Sweep amplitude	Fixed 0.40 rad	$1.5P_{99}(\delta)$, clipped
Pacejka solver	Trust-region [43]	Differential evolution [42]
Tikhonov weight λ	0	0.05, frozen target
Friction warm start	Off	On; D pinned per axle
Update relaxation β	1.00 (undamped)	0.20

Everything not listed is identical between the two, the tire prior and the box constraints included. β is the relaxation of (14); 1.00 is the inherited undamped update. The rollout speed is taken from the measured lap in both arms and is not a difference here (Section VI-A3).

the adaptivity claim a friction sweep would support is not made (Section VIII-B).

2) *Reference trajectory*: Both runs follow the same optimised raceline (Fig. 3), generated with the TUM global race trajectory optimisation package [48] under its minimum-curvature formulation [31]: a closed 5828.2 m lap of 2916 waypoints, reference speed 9.62–25.78 m/s, minimum corner radius 18.7 m ($|\kappa|_{\max} = 0.0536/\text{m}$). Its peak lateral demand is $\max(v_x^2|\kappa|) = 4.98 \text{ m/s}^2$ (0.51 g), roughly 73 % of the 0.70 g this surface sustains. It replaces an earlier trajectory whose 1.22 g demand the admissibility gate of Section VII rejected against the same plant; that regeneration, not a relaxed gate, unblocked the closed-loop runs.

3) *The two configurations*: Two configurations of the same pipeline are run back to back on that trajectory, three timed laps each, with an untimed out-lap. They differ in the eight settings of Table III and in nothing else; vehicle, map, trajectory, controller, data budget, loop count, tire prior and box constraints are shared. Baseline-NN-MSE is the inherited configuration evaluated at full scale on the corrected rollout speed, so the comparison isolates those eight settings rather than reproducing the published pipeline in full; Excitation-S4D-Pinn is the shipped configuration. Since the eight settings move together, what follows is a configuration comparison, not a per-factor ablation, which Section VI-I supplies offline for the four that carry the result. Nor is it a test of this paper's central contribution: the rollout speed is taken from the measured lap in *both* arms, the mean speed under MPC being 22.2 m/s, so Excitation-S4D-Pinn's 15 m/s floor never binds. The excitation correction itself, together with the operating point it ships, is established in Section VI-B.

4) *Provenance of the tire prior*: The prior Φ_p^0 both arms start from, and that the Tikhonov term of (12) pulls toward, is the measured tire of Table II, obtained by fitting (4) to CARLA's own per-wheel telemetry over a dedicated limit-excitation run that reached $\alpha = 0.37 \text{ rad}$, axle $\mu = 1.05$ and 1.50 g. It is not independent of the validation data, since that is the same channel the force and friction results are scored against; the manoeuvre that produced it is one the pipeline is not allowed to run, the raceline peaking at 0.51 g; and, decisively, it was measured on a surface these runs do not use, carrying $D = 1.009/1.002$ and axle slopes of 13.2 kN/rad

front and 13.9 kN/rad rear against the 0.70 and the 17.4 and 16.6 kN/rad this surface realises. Returning it unchanged is a 44 % grip overstatement and a 16–24 % stiffness deficit, so an arm has to move away from that prior, in the right direction, to score well.

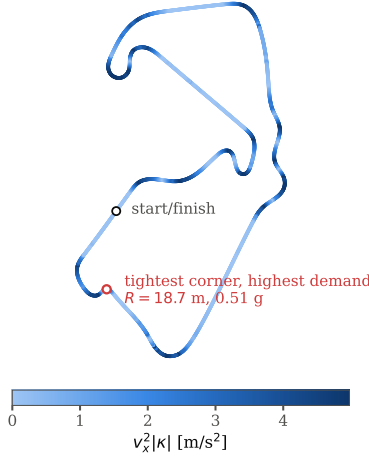
B. Selecting the rollout operating point $(v, \delta_{\text{sweep}})$

This grid is where the rollout configuration the pipeline ships is *chosen*, not an auxiliary check on a choice made elsewhere: the 15 m/s speed floor and the $\delta_{\max} = 0.54 \text{ rad}$ clip of Section V-B are read off Table IV, and the reachable-friction bound (11) is what makes the search one-dimensional rather than a blind scan. The search is run where the answer is known exactly rather than in CARLA, since a CARLA lap cannot be scored against a D nobody knows. A tire with a known peak factor D is rolled out through the synthetic sweep of stage 2 (the shipped rollout, residual network set to zero) and fitted back with the shipped solver, *unregularised* and started from the true coefficients, so everything it loses is lost to excitation. Rollout speed v , sweep amplitude δ_{sweep} and true D are swept, each cell repeated over five noise realisations at the one-step persistence RMSE measured on CARLA runs of this vehicle. The configuration axes are swept one at a time, because μ_{reach} is a function of both; the repetition is over noise rather than the solver because the noise-free fit is exactly seed-invariant here. The selection criterion is the one the pipeline needs: the widest range of true D recovered without railing, over the whole 0.70–1.80 column span, since the tire is unknown when the configuration is fixed.

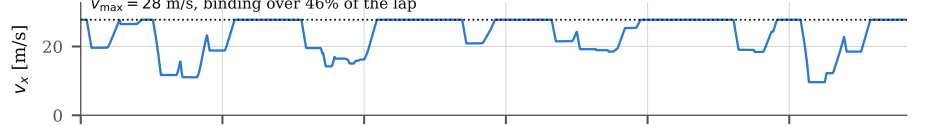
The grid says three things. First, the inherited 4 m/s rollout is uninformative about D at *every* grip level tested: the 0.70, 1.05, 1.40 and 1.80 tires all come back as $\hat{D}$ between 0.50 and 0.59, with two to seven of the eight coefficients on a bound, so the fit reports μ_{reach} and not the tire. Second, the boundary is $\mu_{\text{reach}} \gtrsim D$ and not a speed: at 6 m/s, where $\mu_{\text{reach}} = 0.96$, the 0.70 and 1.05 tires are recovered to within 3 % while the 1.40 and 1.80 tires are 13 % and 30 % short; for μ_{reach} between 1.7 and 4.5 (the $v = 8$ and $v = 13$ rows) every tire is recovered to within 4 % with no systematic railing. The bound predicts where the information runs out, not a step, and it is not a quantity to maximise: at $v = 20 \text{ m/s}$, $\mu_{\text{reach}} = 10.64$, the rollout sits far past saturation for every tire on the grid and $\hat{D}$ re-biases upward, the 0.70 tire returning 0.92. Third, the failure is not the solver's: multi-start, a global solver or a better start point cannot recover a coefficient the data does not contain.

Panel (b) sweeps the other argument of (11) at fixed $v = 15 \text{ m/s}$. Starving the amplitude destroys D just as starving the speed does, so speed is not the governing variable, and the direction of the error is set by the rollout rather than the tire: an under-speeded rollout returns *too little* grip, an under-swept one too much (at $\delta_{\text{sweep}} = 0.06 \text{ rad}$ the 1.05 tire comes back at 1.30 and the 1.40 tire at 1.68, two of the eight coefficients on a bound throughout the under-swept rows). That is the signature of an unidentified parameter, not a biased one. The axes do *not* collapse onto a common μ_{reach} , for the upper-bound reason of Section V-B: at $v = 20 \text{ m/s}$ (11) reads 10.64 against a realised 1.40–2.18. Read against the realised demand both axes behave

(a) raceline, 5.83 km lap, coloured by lateral demand



(b) reference speed profile



(c) lateral demand against available grip

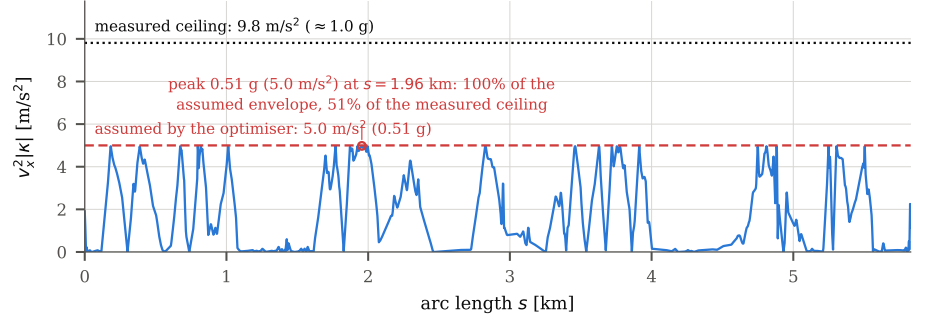


Fig. 3. The optimised raceline used for both runs [48], [31]. (a) Track layout, coloured by lateral demand, with the marked corner ($R = 18.7$ m) the tightest on the lap. (b) Reference speed profile. (c) Lateral demand $v_x^2 |\kappa|$ along the lap against the flat 5.0 m/s^2 envelope handed to the optimiser (dashed) and the grip the vehicle has on this surface (dotted, ≈ 0.70 g, measured). No segment exceeds the plant's grip.

TABLE IV

THE SEARCH THAT FIXES THE SHIPPED ROLLOUT OPERATING POINT: PEAK FACTOR $\hat{D}$ RECOVERED FROM A KNOWN TIRE, SWEEP ALONG BOTH ARGUMENTS OF μ_{reach} . THE SHIPPED $(v, \delta_{\text{sweep}})$ IS SELECTED FROM THIS GRID.

swept	μ_{reach}	$\hat{D}$ (railed of 8) at true D of			
value		0.70	1.05	1.40	1.80
(a) rollout speed v [m/s] swept at $\delta_{\text{sweep}} = 0.40 \text{ rad}$					
4	0.43	0.52 ± 0.036 (1.8)	0.57 ± 0.013 (2.8)	0.59 ± 0.018 (3.8)	0.50 ± 0.001 (7)
6	0.96	0.68 ± 0.001 (1)	1.04 ± 0.014 (1.2)	1.22 ± 0.066 (1.8)	1.26 ± 0.097 (2)
8	1.70	0.69 ± 0.001 (1)	1.02 ± 0.001 (0.8)	1.36 ± 0.002 (0)	1.79 ± 0.025 (1.2)
13	4.49	0.72 ± 0.001 (0)	1.05 ± 0.002 (0)	1.38 ± 0.001 (0.4)	1.77 ± 0.002 (0.4)
20	10.64	0.92 ± 0.004 (1)	1.14 ± 0.005 (0.4)	1.43 ± 0.002 (0)	1.80 ± 0.005 (0)
(b) δ_{sweep} [rad] swept at $v = 15 \text{ m/s}$					
0.03	0.45	0.80 ± 0.019 (2)	1.10 ± 0.041 (2)	1.08 ± 0.182 (2)	1.09 ± 0.397 (2)
0.06	0.90	0.92 ± 0.003 (2)	1.30 ± 0.008 (2)	1.68 ± 0.017 (2)	1.80 ± 0.232 (3.4)
0.15	2.24	0.99 ± 0.004 (2)	1.38 ± 0.003 (2)	1.71 ± 0.019 (2)	1.98 ± 0.009 (1)
0.40	5.98	0.76 ± 0.009 (1)	1.05 ± 0.002 (0)	1.39 ± 0.002 (0)	1.78 ± 0.001 (0)
0.54	8.08	0.83 ± 0.004 (1)	1.08 ± 0.002 (0)	1.38 ± 0.001 (0)	1.78 ± 0.008 (0)

Unregularised, started from the true coefficients. Each cell is $\hat{D} \pm$ the half-width of its 95% t -interval over five noise realisations, with the mean number of the eight coefficients resting on a box constraint in parentheses. The two axes of one rollout are not independent, so each realisation contributes the mean of its own $(\hat{D}_f, \hat{D}_r)$ and $n = 5$.

μ_{reach} is (11) at $L = 1.5334 \text{ m}$; the inherited configuration is the first row of (a). It is the Ackermann upper bound and above $\mu_{\text{reach}} \approx 2$ detaches from what the rollout realises, so a large value means "far past saturation", not a demand the tire ever sees.

alike, so (11) should be used as the pipeline uses it: a cheap screening bound, conclusive when it falls below D , and not a measure of excitation achieved.

The two panels together fix the shipped point. On speed, the admissible band is bounded on both sides: below by railing ($v \leq 6 \text{ m/s}$, which loses the 1.40 and 1.80 tires) and above by the saturation re-bias ($v = 20 \text{ m/s}$, which returns 0.92 for the 0.70 tire). Only the $v = 8$ and $v = 13$ rows recover every tire in the span to within 4%. The 15 m/s floor of Section V-B is placed inside that band and, being a floor, it is the measured lap speed that governs above it. On amplitude, $\delta_{\text{sweep}} \leq 0.15 \text{ rad}$ holds two of the eight coefficients on a bound at every D , while 0.40 rad and 0.54 rad rail nothing and recover every tire; 0.54 rad is therefore taken as δ_{max} , the ceiling of the $1.5P_{99}$ clip, which is what actually selects the amplitude on a given

lap. The grid does not license raising either knob further: past these values the recovery degrades, which is why the pipeline treats (11) as a bound to clear and not a quantity to maximise. All of it is a best case. With the residual set to zero, a Magic Formula is fitted to Magic-Formula data under exactly matched structure, so the selected point is the best setting *on this grid* and not a proven optimum.

C. Cross-run summary

Table V is the complete comparison. Excitation-S4D-Pinn is better on every quantity scored against the plant (peak friction, axle force on both axes, both one-step states, lateral tracking error in every phase, and both error tails) and behind only on QP solve time, by 5% of the RMS and 1% of the worst case, both two orders of magnitude inside the control

TABLE V
CROSS-RUN COMPARISON OF THE TWO CONFIGURATIONS.

Metric	Excitation-S4D-Pinn	Baseline-NN-MSE
<i>Closed-loop tracking (3 timed laps)</i>		
Run duration, wall clock [s]	580.4	554.6
RMS e_y , overall [m]	0.256	0.319
RMS e_y , MPC [m]	0.151	0.224
ITAE e_y [m s^2]	16041	25502
MPC solve, RMS / max [ms]	0.399 / 2.604	0.380 / 2.568
<i>Axle lateral force against per-wheel telemetry</i>		
Front / rear RMSE [N]	27.2 / 33.2	329.6 / 348.4
Front / rear R^2	0.993 / 0.987	-0.160 / -0.461
Front / rear max abs. error [N]	264 / 135	1295 / 1316
<i>Peak friction against the plant's $\mu = 0.70$</i>		
Front / rear μ RMSE, whole run	0.145 / 0.142	1.300 / 1.300
Front / rear μ RMSE, after release	0.078 / 0.078	1.300 / 1.300
<i>One-step state prediction</i>		
v_y / ω RMSE [m/s, rad/s]	0.014 / 0.049	0.156 / 0.104
Skill vs. persistence, v_y / ω	-0.27 / -8.8	-8.18 / -13.9

period. The margins are $9.0\text{--}12.1\times$ on peak friction and axle force and $1.5\times$ on MPC tracking error. The rest of this section reads those rows against the figures; Section VI-I attributes them.

D. Identified tire model

Figure 4 puts both identified curves against the plant's own curve (8) and lists the coefficients.

First, no coefficient of Excitation-S4D-Pinn rests on a box constraint in any of its six accepted identifications; its last accepted set is $B = 12.890/13.746$ inside $[4, 20]$, $C = 1.240/1.248$ inside $[1.2, 2.2]$, $D = 0.768$ on both axles inside $[0.4, 2.0]$ and $E = -2.517/-1.836$ inside $[-3, 1]$. Baseline-NN-MSE puts D on its upper bound of 2.000 on both axles in *every* accepted identification and E_f on its lower bound of -3.0 likewise. The *inner* fits are a separate matter: Excitation-S4D-Pinn's run logs 35 fits that stopped on a bound, 30 of them on C 's lower 1.2 and none on D , which the pin takes out of the fit, against 96 for Baseline-NN-MSE spread over B , C and D . It is the relaxation of (14) that keeps the accepted set off an edge its iterates touch. A peak factor of 2.000 against a plant realising 0.70 is less a 186% parameter error than the box-edge signature of Section III-C: the coefficient is not identified, and no per-coefficient bounds check distinguishes that from a fit.

Second, the linear range is identified rather than inherited. Every slope is (5) for an identified set; for the plant it is the slope fitted to the simulator's own per-wheel forces over $|\alpha| \in [0.005, 0.02]$ rad, all at the static axle loads $F_{z,f} = 1373$ N, $F_{z,r} = 1274$ N, so the three are comparable. The plant realises 17.4 kN/rad front and 16.6 kN/rad rear over this run's load range, below the $\bar{F}_z = 1$ closed-form values of Table II for the reason its first footnote gives. The prior carries 13.2 and 13.9, 24% and 16% low; Excitation-S4D-Pinn returns 16.9 and 16.8, 3.2% low front and 1.4% high rear, removing 87% and 91% of the prior's error. The product $BCDF_z$, the only quantity the rollout fit determines (Section V-D1), is what moves and then holds: the five cycles after the friction gate releases sit within 16.0–16.9 kN/rad, a 5% spread about a plant at 17.4. Baseline-NN-MSE returns 37.9 and 38.4, 118%

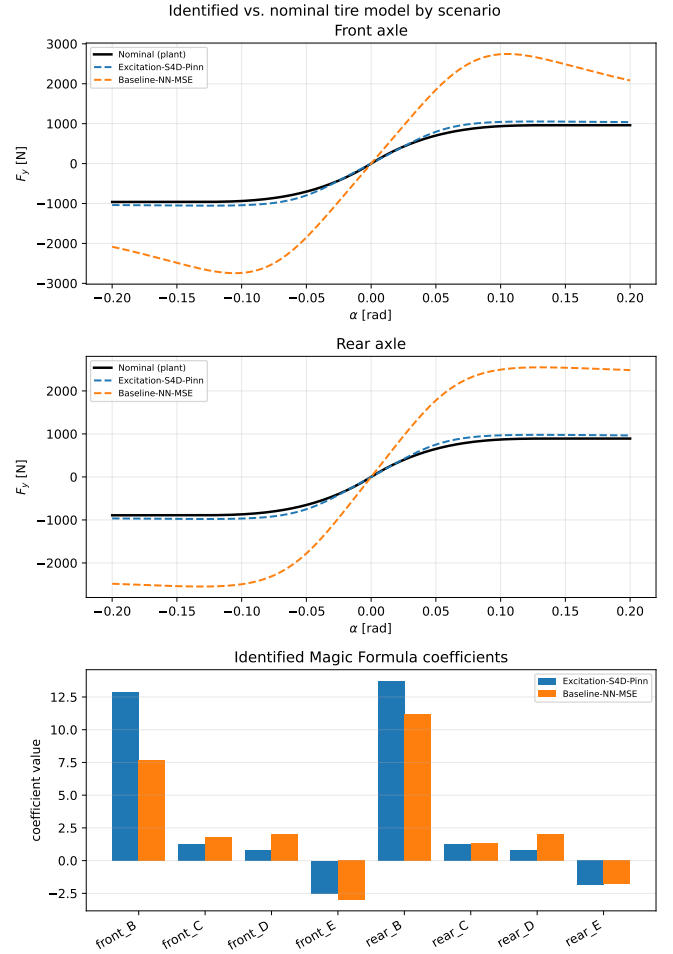


Fig. 4. Identified tire models against the plant. Top and middle: the plant's own curve (8) at the measured static load of each axle (black) against each run's last accepted identification. Bottom: the identified Magic Formula coefficients. Baseline-NN-MSE's accepted peak factor sits on its box constraint $D = 2.0$ on both axles in every cycle, so its curve carries roughly 2.8 times the plant's force at the edge of the plotted range, while Excitation-S4D-Pinn tracks both the plant's initial slope and its peak. All force metrics are axle sums.

and 132% high, which is $D = 2.0$ multiplying a B the fit chose against it.

Third, both sets keep the understeer sign and pass the stability gate of Section VII-A; what separates them is the grip ceiling, which (6) puts at 2.00g for Baseline-NN-MSE against 0.77g for Excitation-S4D-Pinn, a plant sustaining 0.71g and a trajectory demanding 0.51g. The baseline's ceiling is a claim the plant cannot honour by a factor of 2.8; only the identified set is admissible on the quantity the controller consumes. All three readings are possible because the lap now traverses part of the curvature: $P_{99}(|\alpha|) = 2.7^\circ$ and $\max|\alpha| = 5.8^\circ$ against the 8.0° at which the plant's curve peaks. Against the static axle loads the force reaches $\max|F_y|/F_z = 0.71$ front and 0.72 rear, the whole of what the surface offers, but only in transients: at P_{99} the same ratio is 0.49 and 0.50, about 0.70 of the axle's peak force. This is still not a limit manoeuvre, but it suffices for the friction gate to release, which on the $\mu = 1.05$ runs it did not.

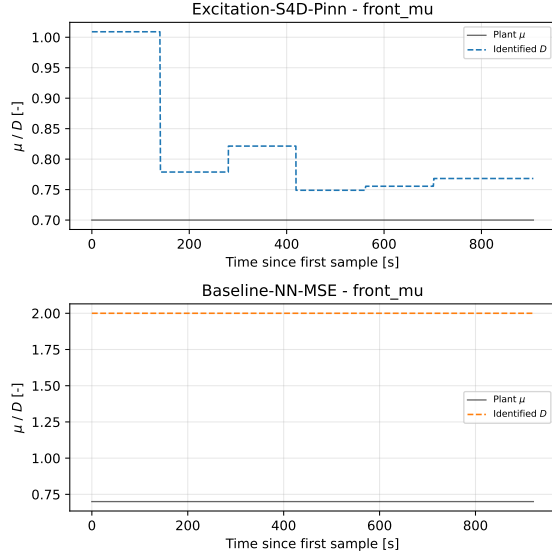


Fig. 5. Peak friction over the run, front axle, one row per configuration: the plant's effective μ (black) against the identified peak factor D published at each accepted identification. The rear axle is identical and is omitted, since D is pinned to the same value on both axles (Section V-E2a). The surface is static, so every departure from the black line is estimator error. Excitation-S4D-Pinn publishes the $\mu = 1.05$ prior while the friction gate is closed, then steps down onto the plant within 11% at the first release and holds there for the remaining five cycles, while Baseline-NN-MSE sits on the $D \leq 2.0$ box constraint for the whole run.

E. Peak friction

Table V reports the error of the identified peak factor against the effective friction the plant reports per wheel, and Fig. 5 its evolution over the run. This is where the pinned- D design of Section V-E2 is directly observable, because the arm starts 44% away from the answer. Excitation-S4D-Pinn publishes six accepted identifications. The first carries $D_f = 1.009$, $D_r = 1.002$: the friction gate rejected that buffer, so the axles held the prior they came in with, the hold-rather-than-fit rule of Section V-E2a and not a fitted value. From $t = 140.6$ s the gate releases on every subsequent cycle and the published peak factor drops to 0.779, 0.821, 0.749, 0.756 and 0.768, front and rear alike, against a plant at 0.700: 10.7% high on average, from a prior that was 44% high. The whole-run μ RMSE is dominated by the first 140 s at the prior; over the 766 s of simulation time after the first release it is 0.078 on both axles. Baseline-NN-MSE returns $D = 2.000$ on both axles in every accepted identification and never leaves the bound, a constant μ error of 1.300, so Excitation-S4D-Pinn is $9.0\times$ better on the whole-run figure and $16.7\times$ better once its warm start has released, a difference worth 1.23 g of claimed grip against a plant that sustains 0.71. Two qualifications belong on that column: the peak factor is *measured* by the brush estimator on the recorded trace and then held, not recovered by the Pacejka fit, which Section V-D1 shows cannot recover it; and the residual 10.7% bias is one-signed, consistent with the brush model's tendency to over-read μ where the curve is not fully traversed.

F. Axle lateral force and one-step state prediction

The identified curve is scored at the measured (α, F_z) against the plant's own lateral force, summed per axle (Table V; Fig. 6(a)). Excitation-S4D-Pinn leads on both axles by an order of magnitude in RMSE and in the error tails; the baseline's negative R^2 means its curve predicts the plant's axle force worse than the mean of that force does. This is the linear-range result seen as force: at $P_{99}(|\alpha|) = 2.7^\circ$ the metric is dominated by cornering stiffness, so the models separate almost entirely on slope. Because D multiplies that slope in (5), Baseline-NN-MSE's railed D is the direct cause of its force error. The comparison is also why the relaxed update of (14) belongs in the method rather than a configuration file: pinning D alone does not produce these numbers, since with $\beta = 1$ the same arm degrades monotonically with iteration count (Section VI-I).

Level 1 of the validation protocol is reported against a persistence baseline (next state equals last measured state) written by the same benchmark node. Excitation-S4D-Pinn leads on both signals, by an order of magnitude on lateral velocity (R^2 0.998 against 0.709) and by a factor of two on yaw rate (R^2 0.894 against 0.514); in Fig. 6(b)–(c) the separation on v_y is in the centre of the distribution, while on ω the two share a centre and differ in tail mass. Neither model beats persistence at the 0.035 s model step, the skill scores being negative for both configurations and both signals, because a one-step metric on a smooth trajectory rewards not moving. One-step accuracy is therefore weak evidence for either tire model, and the gap is read only as confirming the direction the axle-force and closed-loop comparisons establish.

G. Closed-loop tracking

Both runs finish three timed laps with zero emergency halts and the same finite-state sequence, with the handover at $t = 79.1$ s for Excitation-S4D-Pinn (Fig. 7) and $t = 64.0$ s for Baseline-NN-MSE. Under MPC, where the identified model enters the loop, Excitation-S4D-Pinn tracks at 0.151 m RMS lateral error against 0.224 m, a 32% reduction, with maximum 0.683 m against 0.776 m (Fig. 8). ITAE is $\int t|e_y| dt$, neither scale-free nor comparable across runs of different length, so the duration-normalised forms $IAE/T = 0.104$ against 0.157 and $ITAE/(T^2/2) = 0.095$ against 0.166 are reported beside it; all three order the runs alike. The ITAE split confirms that the whole difference sits in the MPC phase: the pure-pursuit phases, which do not use the identified model, give 110 and 107, within 3% of each other and two orders of magnitude smaller. The QP cost is unchanged: RMS and worst-case solve times are two orders of magnitude below the 50 ms control period (Fig. 7) and both slightly in the baseline's favour, by 5% and 1%. The identified coefficients change the model the QP carries, not its size. Lap time is not read as a result: timing runs on the ROS wall clock while the simulator advances its fixed step faster than real time, at a ratio that differed between the runs, so the 5–6% lap-time spread is not a vehicle-performance difference.

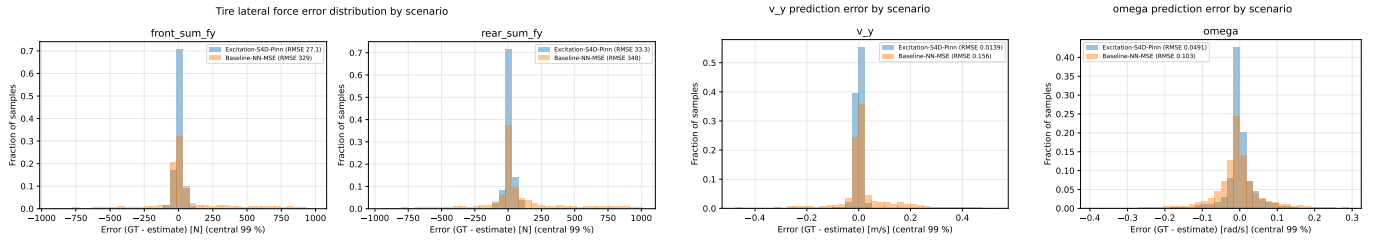


Fig. 6. Error distributions, truth minus estimate. (a) Axle lateral force, shared bin edges over the pooled central 99% and normalised to each run's sample count; both are centred, and Baseline-NN-MSE's distribution is roughly an order of magnitude wider at every quantile (maximum absolute error 1295 N against 264 N front, 1316 N against 135 N rear). (b) One-step lateral velocity, where Excitation-S4D-Pinn is an order of magnitude tighter (0.014 m/s against 0.156 m/s RMSE). (c) One-step yaw rate, where the configurations differ in tail mass rather than centre. Neither model beats its own persistence baseline at the 0.035 s model step.

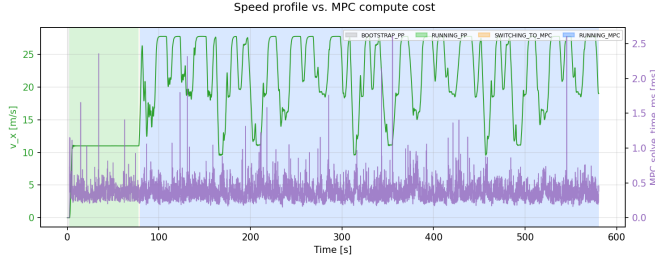


Fig. 7. Speed and solver cost along the Excitation-S4D-Pinn run: longitudinal speed v_x (left axis) against MPC QP solve time (right axis) over the full 580.4 s, shaded by the active controller state. The handover from pure pursuit to MPC falls at $t = 79.1$ s; after it the vehicle cycles the raceline's 9.6–25.8 m/s reference profile while the solve time stays two orders of magnitude below the 50 ms control period. Excitation-S4D-Pinn only; not a comparison between arms.

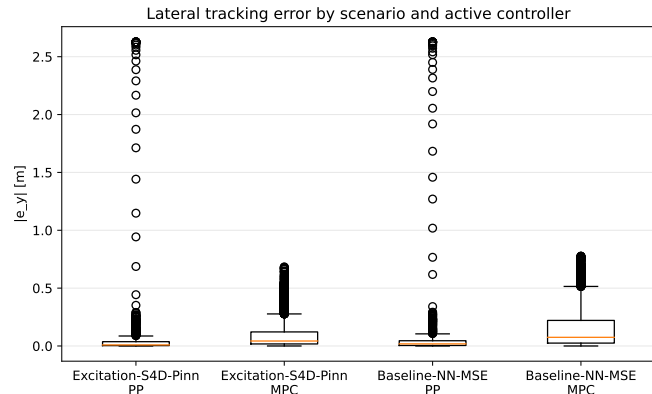


Fig. 8. Absolute lateral tracking error by configuration and active controller. The pure-pursuit boxes cover the acquisition phase before handover and do not use the identified model. The MPC boxes do.

H. Peak-friction warm start on a synthetic plant

The friction warm start is evaluated on a synthetic single-track plant with brush tires, because only a synthetic plant supplies a μ known exactly and an excitation level that can be swept independently of it: a swept-sine steering input for 40 s at 50 Hz with true $\mu = 0.70$, the same effective peak as the CARLA surface, the steering amplitude setting how far up the tire curve the manoeuvre reaches (Table VI).

At 63% utilisation, a level an ordinary lap reaches, the utilisation bound is 37% low while the brush fit is within 0.2%,

TABLE VI
PEAK FRICTION RECOVERED FROM A SYNTHETIC BRUSH-TIRE PLANT (TRUE $\mu = 0.70$), AGAINST EXCITATION.

δ amplitude	Grip reached	Bound $\hat{\mu}$	Brush fit (16)
0.033 rad	45 %	0.312	gated off
0.047 rad	63 %	0.442	0.699
0.060 rad	80 %	0.557	0.699
0.090 rad	100 %	0.700	0.697

Column 2 is the fraction of the available grip the manoeuvre reached, $\max |F_y|/(\mu F_z)$; column 3 is the utilisation bound $\max |F_y|/F_z$, which is what that fraction leaves as a lower bound on μ . The brush fit of (16) instead recovers the peak from the curvature of $F_y(\alpha)$ and is therefore flat in excitation.

The first row is refused by the absolute gate $\max |F_y|/F_z \geq 0.40$ of Section V-E2: at $\mu = 0.70$, 45% utilisation is only 0.31 of F_z . The utilisation bound is reported instead.

and it stays within 0.2% across true $\mu \in \{0.7, 1.05, 1.4\}$ on noise-free data. With odometry and IMU noise injected the error is grip-dependent, the noise being absolute while the available force is not: at $\mu = 0.70$ it is 20% at 63% utilisation and 0.4% at the peak. Below the absolute 0.40 floor on $|F_y|/F_z$ the gate suppresses the fit and reports the bound, so the failure is visible rather than silent. The gate behaves the same way on the CARLA run: it rejects the first buffer, and on the five it accepts returns μ within 11% of the plant at an absolute axle utilisation of 0.54–0.68. That 11% bias against the 0.2% of the synthetic plant is the cost of a real rollout, a reconstructed $I_z \dot{\omega}$ term and a curve whose sustained demand is only about 0.70 of the axle's peak force (Section VI-D), and it dominates the identified peak factor's error.

I. What carries the result

The two arms differ in eight settings at once, so that comparison establishes an ordering and attributes nothing. This subsection attributes it, at the cost of leaving the loop: every number below is an *offline replay* of the identification stage on the five 30 s buffers each arm collected during its own run, scored with the quantity the closed-loop tire-force benchmark itself reports, so these numbers and Table V's are on the same scale.

1) *Damping the co-identification loop*: Table VII sweeps β of (14) at six iterations throughout; two columns are the argument. At $\beta = 1.00$, the inherited undamped update, axle force RMSE is 70.4 and 62.3 N, reached by degrading

TABLE VII
RELAXATION SWEEP. FIVE RECORDED BUFFERS, SIX ITERATIONS,
OFFLINE REPLAY.

β	0.00	0.05	0.10	0.15	0.20	0.25	0.30	0.50	1.00
Front F_y RMSE [N]	92.5	68.3	48.6	33.9	28.0	31.2	36.4	55.6	70.4
Rear F_y RMSE [N]	71.8	49.0	31.3	19.8	19.4	26.0	30.2	38.4	62.3
Front BCD [rad]	7.82	8.82	9.66	10.36	10.88	11.32	11.65	12.75	12.42

$\beta = 1.00$ is the inherited undamped update and $\beta = 0.00$ the pinned prior, i.e. no identification. 0.20 ships.

monotonically from 38.9 and 26.3 N at a single iteration: the loop amplifies its own extrapolation instead of refining the estimate. At $\beta = 0.00$ the coefficients never move from the pinned prior, which is no identification at all, and that is the worst column, 92.5 and 71.8 N. The minimum sits between them at $\beta = 0.20$, 28.0 and 19.4 N, a mean of 23.7 N, and the identified BCD rises with β over the whole damped range (7.82 at $\beta = 0$ to 12.75 at $\beta = 0.50$; the undamped 12.42 is no longer ordered by it), so the damping trades how much of each fit is accepted against how much extrapolation error comes with it. Six iterations are therefore defensible on this evidence, provided the update is damped.

2) *The other three settings:* Pinning the peak factor accounts for most of what remains. Replayed on five buffers of a $\mu = 1.05$ run, letting the fit move D gives a mean $|D - \mu|$ of 0.585 and a front-stiffness spread of 32.9 kN/rad; pinning only the axles whose brush fit was released reduces those to 0.191 and 32.3, and also holding the axles with no released measurement, the shipped rule, to 0.084 and 18.5 kN/rad. What removes most of the error is refusing to let an unmeasured D be fitted at all. Normalising the physics-informed objective is necessary but not sufficient alone: it moves the loss shares from 99.9996% on the dynamics residual to 1.47%, yet applied without the damping it makes the force metric slightly worse (77.1 and 76.2 N against 70.4 and 62.3 N), because an objective that now fits the data supplies a sharper trajectory for the undamped loop to amplify. Two settings deserve a weaker statement than the closed-loop comparison alone would support: the S4D residual is worth 21% on the force metric at matched settings and a single iteration (35.3 N against the baseline MLP's 44.7 N), an offline replay and not a closed-loop result; and the residual *objective* is a wash, `physics_informed` and `plain_mse` being indistinguishable at 23.7 N each once the normalisation and the damping are both in place. We keep it because it costs nothing and is the paper's stated formulation, not because it buys accuracy.

J. Identification cost

One S4D identification, six co-identification iterations on a 30 s lap, takes 20.9 s end to end, inside the 30 s re-identification period. The reduction from the 181.4 s it originally took is exact: no epoch count, iteration count, early-stopping rule or loss weight changed, and the identified coefficients agree to four decimal places. The per-phase breakdown is in the supplementary material.

VII. INTEGRATION WITH THE CONTROLLER

The identified model leaves the pipeline through one interface. A nonlinear MPC path-tracking controller (acados [29], partial-condensing HPIPM [30], horizon $N = 20$, run at 20 Hz) carries the same single-track model as the identifier and consumes Φ_p directly through (4), pushed over a ROS 2 service at the end of each identification cycle. We claim no contribution to the controller design; it appears here as the consumer that defines what an admissible identification is.

A. Admissibility gates

A coefficient set can lie strictly inside every box constraint and still be unusable. We observed two ways, each requiring a gate on a *derived* quantity rather than a coefficient:

- 1) **Grip ceiling versus trajectory demand.** The controller computes $\max_k (v_{x,k}^2 |\kappa_k|)$ over the speed-clamped reference and rejects any identification whose $a_y^{\max}$ from (6) falls below it. On the earlier trajectory the degenerate fit reported a 0.40 g ceiling against a 1.22 g demand and produced 41.89 m of cross-track error at 27 m/s, against 0.03 m for a physically plausible set ($D = 1.5$) on the same trajectory, at the same speed and through the same controller; the same identified set gives 0.16 m at 11.1 m/s and 0.43 m at 15 m/s, the v^2 scaling the Introduction describes. These figures come from an offline closed-loop harness rather than a controlled pair of simulator runs, and are quoted as the observation that motivated the gate, not as a result. The gate is one-sided by construction and catches only a ceiling that is too low, so Baseline-NN-MSE's 2.00 g (Section VI-D) passes it.
- 2) **Understeer sign and critical speed.** A set with $l_f C_f > l_r C_r$ whose v_{crit} falls inside the controller's speed range makes the prediction model open-loop unstable, and the resulting $\rho(A_d) > 1$ is raised to ρ^N in the condensed QP. An observed set gave $C_f = 37031$ N/rad against $C_r = 11691$ N/rad, i.e. $v_{\text{crit}} = 14.5$ m/s at the measured geometry and mass of Table I, inside the controller's speed range on every trajectory in this paper.

A third observation is a controller-implementation detail rather than a gate on the identification: an admissible set can still be stiff relative to the control period, so the prediction model sizes its RK4 sub-step from its own fastest lateral mode, restoring $\rho \approx 1.000$ across the speed range, at a median QP setup-and-solve cost of 0.36 ms against the 50 ms control period.

An identification that passes the gates is a model the controller can be held responsible for; one that fails is reported as unidentified rather than deployed. Two further controller-side changes were required before any identified model could be evaluated at full-scale speeds: reference speeds are clamped at ingest, and the solved state is rolled forward by the measured transport delay before each solve. Both are recorded for completeness and are not identification results.

VIII. DISCUSSION AND LIMITATIONS

A. What generalises, and what does not

The reachable-friction bound (11) is kinematic: it involves no tire parameter, no simulator-specific quantity and no property of the residual model, so it applies to any implementation of the virtual-steady-state-data strategy of [6], in simulation or on hardware. Its practical content is a design rule: the rollout must be configured to demand at least the friction one intends to identify, and where it cannot, D should be reported as unidentified rather than fitted. The diagnostic signature, coefficients resting exactly on box constraints, is likewise generic and should be checked and logged by any bounded tire-fitting routine, being otherwise indistinguishable in the output from a successful fit. The two structural properties of Section V-D generalise on the same grounds: the BCD degeneracy follows from the quasi-steady force recovery (7) defining the virtual-data step, and the absence of contraction follows from stage 4 alone, the remedy reducing to the inherited behaviour at $\beta = 1$. What does not generalise is the value 0.20, a property of this vehicle's extrapolation error; what does is that $\beta = 1$ is not defensible and $\beta = 0$ worse still. The insufficiency of per-coefficient box constraints is also platform-independent: both failures we encountered, a set that satisfies every box constraint while degenerate and one that satisfies every box constraint while pairwise oversteering and unstable inside the controller's operating range, are properties of the parameterisation rather than of CARLA.

Three results are properties of this plant and must not be read as statements about tires: the numerical tire values, since the plant is a PhysX wheeled-vehicle model and agreement with Table II shows only that the identifier recovers the plant it observes; the gap between configured wheel friction and realised peak axle friction (1.0 against 0.70), an artefact of the server multiplying the configured coefficient by the road surface's own, which we report because an identifier tuned to converge on the configuration file's number will converge on the wrong one; and the 0.76 achieved/commanded steering ratio, where the lesson (identify against the achieved actuator state) is general but the magnitude is not. Sagmeister *et al.* [26] show sensitivity to model fidelity is largest near the acceleration limits, where this study operates, and R-CARLA [25] exists because CARLA's native dynamics are a known weak point for racing-grade work, so hardware replication is necessary before any claim about identification accuracy on a real vehicle.

B. Limitations

- 1) **One run per configuration, so no metric in Table V carries an uncertainty.** Every value is a point estimate from a single 580.4s and a single 554.6s run. The 0.151 m against 0.224 m RMS tracking difference in particular has no confidence interval and no evidence that it exceeds run-to-run variation, which is not measured here; the order-of-magnitude force and friction margins are large enough that such variation is unlikely to reverse them, but that is an argument, not a measurement.

- 2) **Single surface, vehicle and trajectory.** That the surface differs from the one the prior was measured on makes the identification a real result rather than a restatement of the prior, but it is *not* a demonstration of adaptivity: the identifier is never asked to follow a *change* in friction. The runtime friction override and the step and decay schedules the scenario file already carries support that experiment, and we do not perform it here.
- 3) **The friction estimator's excitation sweep is synthetic** (Table VI). On the CARLA runs the warm start is exercised at the one excitation level the lap supplies, returning μ 11 % high against the 0.2 % of the synthetic plant, so the excitation dependence of that bias is not mapped against the simulator's per-wheel forces. The published peak factor is exactly this estimate, so the bias passes through undamped.
- 4) **The rear axle is not identified on this trajectory** and the front-axle μ is applied to both (Section V-E2a). The quasi-steady argument making $\mu_f = \mu_r$ holds on a homogeneous surface and would not survive a split- μ one, which this sensor set cannot detect in the first place.
- 5) **The two configurations differ in eight settings at once** (Table III), so Section VI-C attributes nothing to the residual architecture, the objective, the integrator, the sweep bound, the solver, the regularisation, the warm start or the relaxation individually. Section VI-I does attribute four of the eight, but by offline replay on recorded buffers, so its ordering is not a closed-loop ordering, and the integrator, the sweep bound and the regularisation are not separated at all. The S4D figure there (21 % on force RMSE at matched settings) has the same status: we report it as a measured ablation and not as a reproduction of the closed-loop S4 improvement of [20]. The excitation-aware rollout, this paper's central contribution, is active in *both* arms, so the closed-loop comparison is not evidence for the identifiability correction; the evidence for that, and the search that fixes its operating point, is the sweep of Section VI-B.
- 6) **The physics-informed objective buys nothing measurable here.** Once its units are reconciled with the data term and the loop is damped, it is indistinguishable from a plain one-step MSE on this manoeuvre. It is retained because it is the paper's stated formulation and costs nothing, not because it is shown to help.
- 7) **Longitudinal dynamics are out of scope.** As in the baseline, only lateral dynamics are identified. Combined slip and load transfer are neglected, and the simulator's longitudinal wheel force channel is drivetrain torque rather than a contact-patch force, so it is unusable as ground truth for a longitudinal extension without further work.

IX. CONCLUSION AND FUTURE WORK

This paper studied the identifiability of a learning-based on-track Pacejka identification pipeline [6] at full vehicle scale, using a CARLA simulation architecture that exposes

the plant's own per-wheel forces as ground truth. Transplanted unchanged from a 1:10 platform, the pipeline fails in a specific, diagnosable and correctable way: a loss of excitation created by its own virtual-data step, which no change of solver, bounds or tuning removes, and whose reachable friction depends on the rollout configuration and not the tire. Two further properties of the loop follow from the same analysis. The fit's residual depends on B , C and D only through their product, because stage 3 reconstructs its forces from the rollout's own quasi-steady yaw balance and so observes nothing but the lateral acceleration the previous coefficient set produced; the peak factor is therefore not recoverable from the rollout at any excitation and must be measured on the recorded trace and held. And because each fit is handed the trajectory its own previous answer generated, the loop is a positive feedback path rather than a contraction, so stage 4 is reformulated as a relaxed fixed-point iteration.

The correction comes with an extrapolation bound on the steering sweep, a frozen regularisation target, identification against the achieved rather than the commanded road-wheel angle, measured mass and geometry, a friction warm start identified from the curvature of the axle force curve and pinned for the loop, a relaxation $\beta = 0.20$, and a dynamics residual normalised to the units of the data term. Together these produce accepted identifications with no railed coefficient and the correct understeer sign, at the margins Section VI reports. Those numbers are an identification and not a preserved prior, with three qualifications: the peak factor is identified by the brush estimator on the recorded trace, not by the Pacejka fit, which cannot identify it (Section V-D1); one surface is not adaptivity, which needs a friction *change* the runs do not contain; and each configuration is a single run, so the closed-loop margins are observations rather than estimates with a confidence interval. We further argue that per-coefficient box constraints are not a sufficient admissibility criterion for handing a model to a model-based controller, and place gates on derived physical quantities at that interface instead.

A. Future work

- 1) **Sweep the friction rather than merely mismatching it.** These runs put the prior 44 % away from a *static* surface, showing the pipeline moves on evidence but not that it tracks; the step and decay schedules already in the scenario file are the intended vehicle, and the strongest available argument for identifying on track at all.
- 2) **Run the ablations of Section VI-I in the loop.** The relaxation sweep, the D -pinning comparison and the S4D and objective ablations are offline replays, enough to attribute the result but not to confirm the ordering survives closed-loop coupling. Each is a scenario-file row.
- 3) **Excite the peak, and design for it.** The lap reaches $P_{99}(|\alpha|) = 2.7^\circ$ against a plant whose curve peaks at 8.0° ; that is enough for the friction gate to release and is also where the brush estimator's 11 % bias comes from. Equation (11) inverts into a requirement on the manoeuvre, so a controller that deliberately inserts

bounded slalom or ramp-steer excitation when the identifier reports insufficient slip-angle coverage would move the pipeline toward active, safety-bounded experiment design [49].

- 4) **Hardware replication.** The decisive test of every claim in Section VIII is a physical Formula Student vehicle. The identification code is unchanged between simulation and hardware; what changes is the availability of ground truth, and that is what makes the simulation study a necessary precursor rather than a substitute.

APPENDIX A

ASSUMPTIONS AND REPRODUCIBILITY NOTES

The following are assumptions or design choices rather than measured facts, listed so a reader can judge how far each result depends on them.

A1. Yaw inertia. $I_z = 51.1 \text{ kg m}^2$ is carried in the model file and is *not* measured. It enters (2) directly, so it scales the yaw-rate residual the network is trained on. It can be discharged by applying a known yaw moment and fitting I_z from the measured $\dot{\omega}$, or by reading the actor's inertia tensor.

A2. Centre-of-gravity split. $l_f = 0.738142 \text{ m}$, $l_r = 0.795362 \text{ m}$ are taken from the controller configuration. The measured static load split (51.2/48.8) implies a marginally different split than the geometry does (51.9/48.1), a 1.4 % disagreement, below the level at which any conclusion here would change.

A3–A4. CG height and steering sign. $h_{cg} = 0.3 \text{ m}$ is carried but not measured. The single-track identification model neglects load transfer, following the baseline, and its normal loads are static; the only consumer of h_{cg} is the longitudinal transfer term of the friction warm start (Section V-E2), where it moves the axle loads by up to 14 % at the 0.35 g the fit admits. `feedback/steering_angle` is published in degrees and treated as the front road-wheel angle, positive left, verified against the front-left wheel joint state to within 0.5 %; left and right road wheels are not distinguished.

A5. Steady-state force recovery. Equation (7) is not a force measurement; being fully determined by $v_x \omega$ it misattributes transients, and it is applied only to the *synthetic*, quasi-steady rollout. Its adequacy was checked on measured data against the simulator's own axle force (correlation 0.993, slope 0.94).

A6. The residual network is not the plant. The corrected model is queried by the rollout at inputs the network has only partially seen. The extrapolation bound of Section V-C limits but does not eliminate this: the sweep still reaches $1.5 \times$ the observed 99th percentile of $|\delta|$, a choice that is empirical rather than derived.

A7. Simulator determinism and timing. CARLA runs in synchronous mode with a 0.033 s physics step; sim time advances faster than wall time, at a ratio set by the host load and therefore not identical across runs (1.56 and 1.66 on the two runs of Section VI). All identification and force-validation data is stamped from the simulation clock, while the controller node stamps its own metrics on the wall clock; the two are never mixed inside one metric (Section VI-A).

A8. Software configuration of record. ROS 2 Humble, Python 3.10, PyTorch with CUDA on an RTX 2080-class

TABLE VIII
ROS 2 INTERFACE USED BY THE IDENTIFICATION AND VALIDATION
PIPELINE. NAMESPACE IS /SIM UNLESS MARKED ABSOLUTE.

Topic	Type	Role
/odom	nav_msgs/Odometry	State v_x, v_y, ω ; twist is body-frame
feedback/ steering_angle	std_msgs/Float32 (deg)	Achieved road-wheel angle δ
feedback/imu	sensor_msgs/Imu	a_x, a_y for the friction warm-start
feedback/ tire_forces	sim_manager_msgs/ TireForces	Per-wheel $\alpha, F_z, F_y, \mu, \omega_w, \kappa$: ground truth
feedback/ vehicle_physics	std_msgs/String (JSON)	Live wheel parameters of (8)
feedback/speed	std_msgs/Float32	Forward speed echo
feedback/motors	std_msgs/String (JSON)	Per-wheel speed/torque/brake
/drive	ackermann_msgs/ AckermannDrive Stamped	Control command in
control/ tire_friction	std_msgs/Float32	Runtime grip override
/clock (<i>abs.</i>)	roscpp_msgs/ Clock	Simulation time base

GPU; the bridge is a separate overlay; CARLA server 0.9.16 / Unreal Engine 4.26; acados 0.5.5 with partial-condensing HPIPM. The identifier's acquisition timer runs at 50 Hz (the controller runs at 20 Hz, a 50 ms period) with a 30 s buffer, six co-identification iterations, 200 training epochs, full-batch Adam at 5×10^{-4} , and $T_s = 0.035$ s. `nn_train` seeds PyTorch and NumPy at 42; `pacejka_solver.seed` is null in both shipped parameter sets, so the reported identifications are reproducible in distribution but not bit for bit, while the sweep of Table IV is seeded explicitly and is bit-reproducible.

APPENDIX B SIMULATOR-ROS 2 INTERFACE

Feedback path

Table VIII lists the interface consumed by the identifier and its validation.

a) *Per-wheel ground truth:* `feedback/tire_forces` carries CARLA's own `WheelTelemetryData` in wheel order [FL, FR, RL, RR]. No analytic tire model exists inside the bridge, so nothing on this topic can disagree with the physics the vehicle is driven by. Four measured properties bound what may be read off it. (i) Only `lateral_force` is force ground truth; it is the channel every validation in this paper uses. (ii) `longitudinal_force` is a friction-capacity report rather than a measurement, tracking μF_z and correlating with ma_x at 0.04, and `wheel_torque` is an exact identity of it. (iii) Below 0.5 m/s the publisher zeroes slip angle and both forces while the normal load stays live, so the identifier gates on the all-zero pattern as well. (iv) The slip angle needs no sign change across the boundary, verified rather than assumed.

Figure 9 states the conventions the rest of the paper assumes. CARLA inherits Unreal's left-handed world while every ROS 2 topic follows REP-103 [50] and the identification equations are written in the body frame with α and F_y positive to the left, the SAE-style axis convention [51], [4]. The bridge is a per-channel conversion rather than a single transform, and

the trap is that the tire telemetry's slip angle needs *no* sign change, because the frame flip and CARLA's own convention cancel: an error there is invisible in the states and inverts the identified curve, so it was checked against a slip angle rebuilt from the chassis state in a steady 4 m/s turn, equal in magnitude to within 2 % front and 15 % rear.

APPENDIX C REFERENCE IMPLEMENTATION

The identification package, the ROS 2 bridge and the controller stack used in this paper are available as On-Track-SysID, extended from [6] (upstream: <https://github.com/ForzaETH/On-Track-SysID>), CARLA_ASU_BRIDGE (https://github.com/ebrahimabdelghfar/Carla_ASU_Bridge) [37], and the accompanying MPC and benchmarking packages, released together with the scenario definitions, the exported result CSVs and the figure and sweep generators at https://github.com/ebrahimabdelghfar/Ebrahim_Master_Thesis_repo/tree/feat-follow-premade-racing-track.

ACKNOWLEDGMENT

The ON-TRACK-SYSID pipeline studied here originates from the ForzaETH Race Stack and the work of Dikici *et al.* [6], released under an open-source licence; this paper reports a port, a correction, and an extension of that work rather than a reimplement from scratch.

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CONFLICT OF INTEREST

The authors declare no conflict of interest.

ETHICS

All results in this paper are obtained in simulation. No experiment involved human subjects, personal data, or the operation of a physical vehicle, so no human-subjects approval and no vehicle-safety protocol were required.

DATA AND CODE AVAILABILITY

The identification package, the ROS 2 bridge, the MPC and the benchmarking packages, the scenario definitions, the exported result CSVs behind every figure and table of Section VI, and the sweep generator behind Table IV are in the repositories listed in Appendix C; the top-level project repository is https://github.com/ebrahimabdelghfar/Ebrahim_Master_Thesis_repo/tree/feat-follow-premade-racing-track. ON-TRACK-SYSID is an extension of the ForzaETH release of [6] and is distributed under the MIT licence it declares in its package manifest; CARLA_ASU_BRIDGE is GPL-3.0. The two are used as separate ROS 2 overlays communicating only over ROS 2 topics and services, and neither links against the other, so no combined-work obligation arises between them; the identification and control packages in this work are released under the MIT terms the ON-TRACK-SYSID upstream requires.

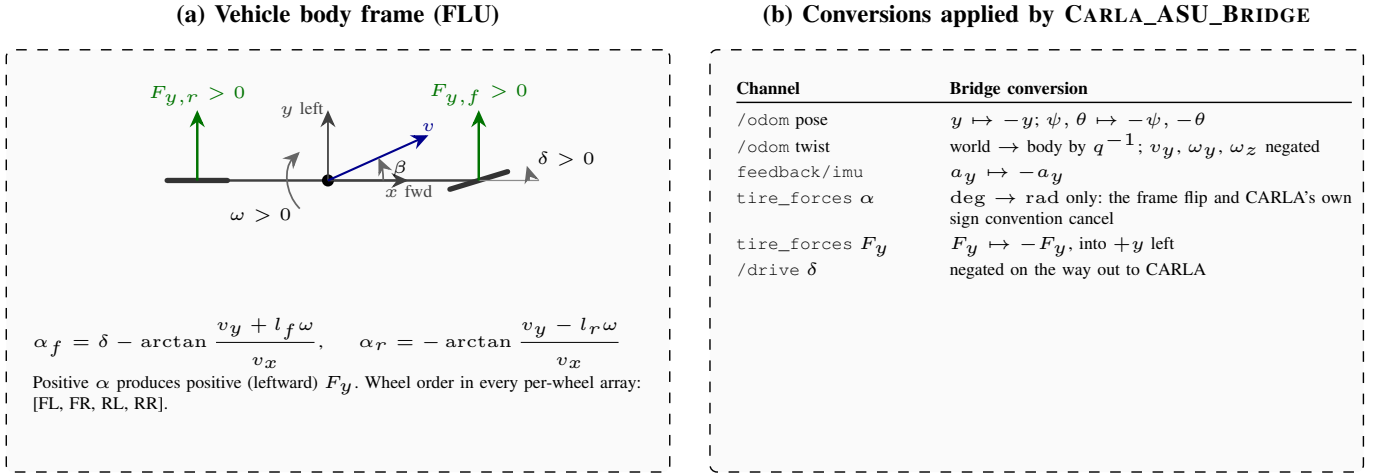


Fig. 9. Frame and sign conventions across the simulator–ROS 2 boundary. (a) All identification quantities are defined in the body frame: x forward, y left, ω and δ positive counter-clockwise, and slip angle and lateral force positive to the left, so that a positive α produces a positive F_y [51]. (b) The conversion applied per channel. The slip angle of the tire telemetry is the one exception that needs no sign change, because the frame flip and CARLA's own convention cancel.

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